

RESEARCH ARTICLE

Environmental Filtering Weakens Latitudinal Diversity Gradients in Soil Biodiversity Worldwide

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ABSTRACT

Aims: The latitudinal diversity gradient (LDG), a foundational pattern describing increasing biodiversity towards the Equator, has been extensively documented for aboveground organisms. However, whether this pattern holds for belowground biodiversity, the largest component of Earth's biodiversity remains unclear.

Location: Globe.

Time Period: Present.

Major Taxa Studies: Soil Archaea, Bacteria, Fungi, Protists, and Invertebrates.

Methods: We put together two data collections (amplicon and metagenomics) encompassing over 5,000 soil samples published online across multiple trophic levels, including archaea, bacteria, fungi, protists, and invertebrates.

Results: Our results demonstrate that multitrophic soil biodiversity deviates substantially from the classical LDG, with most taxa showing no decline in diversity with increasing latitude. We further found that local-scale edaphic properties (e.g., soil carbon, nitrogen, and pH) and global-scale climatic variables (e.g., rainfall and temperature), largely override latitude in explaining soil biodiversity variations. Multi-group structural equation modelling indicated that local diversity was primarily governed by regional diversity and soil nitrogen availability, suggesting that nutrient supply mediates spatial variation in community assembly across regions.

Conclusions: Our findings challenge the universality of the classical LDG pattern and advance our understanding of the mechanisms shaping global soil biodiversity patterns.

1 | Introduction

Soil biodiversity, which constitutes the majority of Earth's biodiversity, is critical for ecosystem stability and functionality, thereby contributing to the achievement of the United Nations

Sustainable Development Goals (Averill et al. 2022; Banerjee and van der Heijden 2023; Crowther et al. 2024; Guerra et al. 2020). The latitudinal diversity gradient (LDG) is one of the well-established ecological patterns that posits higher number of species (i.e., biodiversity) at the Equator compared to the poles

(Hillebrand 2004; Willig et al. 2003). This gradient has been widely documented across a wide range of aboveground taxa, including mammals (Marcot et al. 2016), birds (Duchêne and Cardillo 2015), and insects (Condamine et al. 2012). However, the ubiquity of this pattern in soil organisms, that is, the largest portion of biodiversity of the planet (Anthony et al. 2023), remains debated.

Previous studies have significantly advanced our understanding of the global distributions of various soil communities (Guerra et al. 2022), such as bacteria (Fierer and Jackson 2006; Ji et al. 2022), fungi (Tedersoo et al. 2014), protists (Oliverio et al. 2020), nematodes (van den Hoogen et al. 2019), and earthworms (Phillips et al. 2019), and resources such as the Global Soil Biodiversity Atlas have offered valuable insights into the worldwide distribution and importance of soil life (Orgiazzi et al. 2016). Despite these efforts, considerable controversy and uncertainty remain about whether soil microbial and faunal biodiversity consistently follow the LDG. For example, while some studies report classic LDG for certain taxa like fungi and bacteria (Hyde et al. 2016; Ji et al. 2022), others find weak, non-significant, or even inverse relationships (Tedersoo et al. 2014, 2012). These inconsistencies may arise partly from the distinct environmental drivers governing different taxa: soil pH has emerged as a dominant predictor of bacterial diversity (Bahram et al. 2018; Fierer and Jackson 2006), whereas fungal diversity is more strongly linked to climate variables such as temperature (Tedersoo et al. 2014), and soil faunal diversity is often associated with organic matter availability and habitat structure (Phillips et al. 2019; van den Hoogen et al. 2019). Such taxon-specific responses complicate generalizations about soil biodiversity–latitude relationships. Moreover, most existing studies have focused on single taxonomic groups (e.g., Fierer and Jackson 2006 for bacteria; Tedersoo et al. 2014 for fungi; van den Hoogen et al. 2019 for nematodes). Although a few studies have begun to explore biodiversity across multiple trophic levels simultaneously (Bahram et al. 2018), a comprehensive understanding of how global soil biodiversity conforms to LDG remains elusive (Zhang et al. 2020). Importantly, these taxon-specific inconsistencies suggest that latitude itself may not be the primary driver of soil biodiversity; rather, the underlying environmental conditions that covary with latitude, such as soil properties and climate, may play a more direct and dominant role. Cross-taxon comparisons are essential, as taxon-specific traits such as body size, dispersal capacity, and physiological tolerance may mediate responses to environmental gradients (Martiny et al. 2006).

Biodiversity patterns are inherently scale-dependent. Following Whittaker's conceptual framework (Whittaker 1972), diversity can be partitioned into alpha (local richness), beta (community turnover), and gamma (regional richness). Local diversity is mainly influenced by fine-scale and rapid ecological processes such as interspecific competition, dispersal limitation, and environmental selection, whereas regional diversity patterns are largely shaped by broader evolutionary and historical forces, including speciation and geographic isolation (Wang et al. 2015). These three indices are interconnected, capturing complementary aspects of biodiversity organization; thus, such cross-scale comparisons are essential for a holistic understanding of biodiversity distribution (Mittelbach and McGill 2019). However, most previous studies have focused on alpha diversity,

leaving latitudinal trends in beta and gamma diversity largely unexplored for soil organisms (Bahram et al. 2018; Oliverio et al. 2020).

Numerous hypotheses have been developed to explain the LDG (Mittelbach et al. 2007; Pianka 1966). Key macroecological theories include the environmental heterogeneity hypothesis, which predicts that regions with greater habitat variability support higher species diversity (Stein et al. 2014), and the niche conservatism hypothesis, which posits that species retain ancestral environmental preferences, resulting in higher diversity in regions with longer evolutionary history (Wiens and Donoghue 2004). However, these theories were developed primarily for plants and animals, and their applicability to soil organisms remains largely untested (Martiny et al. 2006; Shade et al. 2018). Several lines of reasoning suggest that soil organisms may deviate from the classical LDG. First, many soil organisms possess enormous population sizes, rapid generation times, and high dispersal capacity, which may decouple their distributions from historical biogeographic constraints and instead make them more responsive to contemporary environmental conditions (Martiny et al. 2006; Shade et al. 2018). Second, the environmental conditions that most strongly influence soil biodiversity, such as soil pH, nutrient availability, and moisture, do not vary monotonically with latitude; rather, optimal combinations of these factors may occur at mid-latitudes, where moderate temperatures, adequate precipitation, and diverse soil substrates create favourable niches for a wide range of taxa (Delgado-Baquerizo et al. 2018; Fierer and Jackson 2006). Third, temperate regions often exhibit pronounced environmental heterogeneity driven by seasonal fluctuations and topographic complexity, which may enhance habitat diversity and promote higher regional species pools (Stein et al. 2014).

Here, we independently analyse two complementary global collections of soil samples published online to test whether the biodiversity of individual soil taxa, including archaea, bacteria, fungi, protists, and invertebrates, conforms to the classic LDG across spatial scales, and to identify the key environmental drivers shaping their global distributions. The first database comprises the surface soil collection, including 615 composite soil samples from 151 sites across major biomes in Guerra et al. (2022). The second database encompasses a metagenomic data collection from 4856 samples spanning diverse biomes worldwide. We test two main hypotheses: (H1) most soil taxa will deviate from the classical LDG, with diversity peaking at mid-latitudes rather than near the equator; and (H2) this deviation is driven by environmental filtering, whereby local-scale soil properties and global-scale climatic variables override latitude as the primary predictors of soil biodiversity. By simultaneously examining alpha, beta, and gamma diversity, which capture complementary aspects of local community assembly, spatial turnover, and regional species pools, respectively (Whittaker 1972), across multiple trophic groups, we provide a comprehensive, multi-scale perspective on the biogeography of soil biodiversity, offering new insights into the ecological and environmental processes governing its global distribution and testing the broader applicability of macroecological theories to the hidden majority of life on Earth. Such insights are urgently needed in the Anthropocene, where rapid environmental changes

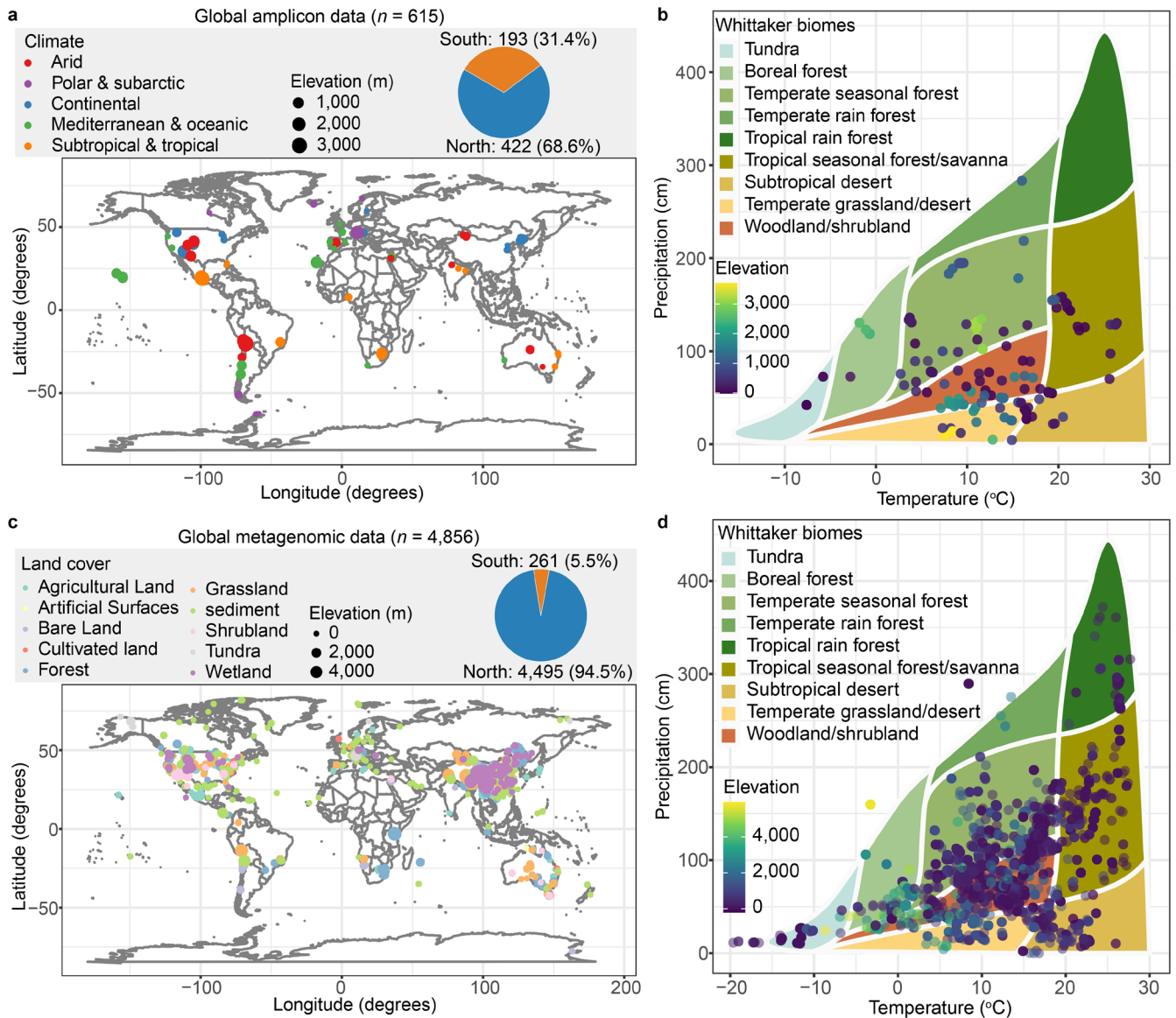


FIGURE 1 | Sample locations in this study. (a) Global map showing the geographic locations of the 615 composite samples (global amplicon data), sourced from Guerra et al. (2022). (b) Whittaker biomes plot highlighting the major biomes of the amplicon data. (c) Global map showing the geographic locations of 4856 composite soil samples (global metagenomic data). These metagenomic data were sourced from the Sequence Read Archive (SRA) database and span major global biomes. (d) Whittaker biomes plot depicting the major biomes of the metagenomic data included in this study.

are profoundly altering ecosystems and reshaping the relationships between soil organisms and their environment (Crowther et al. 2019).

2 | Materials and Methods

2.1 | Study Sites and Datasets

Global amplicon data: Environmental and amplicon sequencing data were obtained from Guerra et al. (2022), with raw sequencing data further accessible through Delgado-Baquerizo et al. (2019) and Eldridge et al. (2023). This dataset comprises a soil collection of 615 composite topsoil samples, with surface soil samples collected at depths up to 10 cm (~0–10 cm), across 151 sites from 23 countries (Figure 1a) in the field surveys linked with Delgado-Baquerizo et al. (2019) and Eldridge et al. (2023). This database encompasses a

diverse range of biomes and environmental conditions (Figure 1b, Figure S1). Detailed soil sampling, DNA extraction, and sequencing procedures are described in the original publications; here, we reanalysed the raw data to address our specific research questions regarding LDGs. Sequencing was done following a standardized protocol in the same research facility in Colorado, USA. Amplicon sequencing of bacterial 16S rRNA and eukaryotic 18S rRNA genes utilized the universal primer sets 515F (5'-GTGCCAGCMGCC GCGGTAA-3')/806R (5'-GGACTACHVGGGTWTCTAAT-3') and Euk1391f (5'-GTACACACCGCCCGTC-3')/EukBr (5'-TG ATCCTTCTGCAGGTTACCTAC-3'), respectively (Amaral-Zettler et al. 2009; Ramirez et al. 2014; Stoeck et al. 2010). Raw sequencing reads were processed following the pipeline of Guerra et al. (2022), which integrates QIIME (Caporaso et al. 2010), USEARCH (Edgar 2010), and UNOISE3 (Edgar 2016) methodologies. Cleaned reads were clustered into zero-radius operational taxonomic units (zOTUs; also known as phylotypes) with 100%

similarity. Both bacterial and eukaryotic taxonomic tables were sourced from Guerra et al. (2022). Reads were rarefied to an even depth for downstream analyses: Bacteria (5,000 reads), fungi (1,000 reads), protists (800 reads), and invertebrates (100 reads). A percentage of the samples did not pass rarefaction; consequently, downstream analyses were based on 578, 578, 569, and 526 samples for bacteria, fungi, protists, and invertebrates, respectively. These whole community rarefied zOTU tables were then used to further determine richness of soil organisms within bacteria (e.g., Actinobacteria), fungi (e.g., Basidiomycota), protists (e.g., Cercozoa) and invertebrates (e.g., Nematoda), respectively. Prior to rarefaction, non-target sequences, including Vertebrate, plants, chloroplasts, mitochondria, and unassigned taxa, were removed from the dataset.

Global metagenomics data: Publicly available soil metagenomic data from 4856 samples were obtained, with 4451 samples retrieved from the Sequence Read Archive (SRA) and 405 from the in-house dataset reported by Ma et al. (2023), representing diverse global biomes and environmental conditions (Figure 1c,d, Figure S2). Soil samples in this published collection are mostly coming from surface soils. Sample selection and all subsequent bioinformatic processing were performed specifically for the present study. Metagenomic datasets from major ecosystems (e.g., forests, grasslands, and wetlands) were selected based on a file size of ≥ 3 Gb to ensure sufficient sequencing depth for reliable taxonomic annotation and diversity estimation. Metadata were collected to define biome types, and when information was incomplete, biome types were inferred using locations from GlobeLand30 (Jun et al. 2014) and Google Maps. Raw reads were processed using Trimmomatic v2.39 (Bolger et al. 2014) for adaptor trimming and filtering out short (< 50 bp) and low-quality reads (< 20 bases). Assembled reads per sample were processed with MEGAHIT v1.2.9 (Li et al. 2015), with a minimum contig length of 50 bp (k-step = 10; k-min = 27). To ensure high sequence quality, only contigs ≥ 10 kb were retained for downstream analyses. Clustering was performed using an ANI pipeline with a 95% average nucleotide identity and 85% alignment fraction (Nayfach et al. 2021). Contig sequences were annotated using CLARK against the GTDB database (<http://gtdb.ecogenomic.org/>), with identification thresholds of > 0.6 and gamma scores > 0.03 (Ounit and Lonardi 2016).

The two databases were analysed independently, as amplicon and metagenomic approaches differ substantially in sequencing strategy, taxonomic resolution, detection biases, sample size, and geographic coverage. Accordingly, comparisons between the two datasets were used only as a qualitative consistency check for broad latitudinal patterns in overlapping high-level taxa (e.g., dominant bacterial phyla). We therefore emphasize directional concordance and the overall shape of diversity–latitude relationships, rather than absolute diversity values, effect sizes, or exact peak latitudes, which are not directly comparable across methods. The goal of using these two independent databases is to provide a general picture of the responses of contrasting soil organisms to latitude.

2.2 | Environmental Predictors

Environmental data was extracted from Guerra et al. (2022). In brief, geographic information, including longitude, latitude, and

elevation, as well as ecosystem types (e.g., forest, grassland, and shrubland), was recorded for each site. Climatic variables, including mean annual temperature (MAT), mean annual precipitation (MAP), precipitation seasonality (PSEA), and temperature seasonality (TSEA), were obtained from WorldClim version 2 (<https://www.worldclim.org/>) at a 30 arc-second resolution. For the amplicon dataset, soil physicochemical properties were directly measured using standardized protocols. Soil texture, specifically defined as clay percentage, was determined using standardized methods (Kettler et al. 2001). Soil pH was measured using a pH meter in a suspension of soil and water, with a mass-to-volume ratio of 1:2.5. Missing pH data ($\sim 5.5\%$ of samples) was estimated using site-level pH information. Soil carbon (C) and nitrogen (N) were quantified using a 2400 II CHN elemental analyser (PerkinElmer, Waltham, MA, USA), and the C/N ratio was calculated from these values. For the global metagenomic database, where direct soil measurements were not available, soil variables were retrieved from the SoilGrids database (<https://soilgrids.org/>) at a 250 m resolution for topsoil layers. Net primary productivity (NPP) data with a spatial resolution of 0.1° were obtained from the MODIS MOD17A3H annual NPP product available through NASA Earth Observations (NEO, <https://neo.gsfc.nasa.gov>).

2.3 | Statistical Analyses

Diversity metrics: We focused on the most abundant phyla or sub-phyla within each major taxonomic group to capture the dominant community structure of the microbial and faunal assemblages. Specifically, we included the top ten most abundant phyla for bacteria, archaea, and protists, seven dominant phyla for fungi, and four dominant phyla for invertebrates. Following Whittaker's classical framework for analysing diversity across spatial scales (Whittaker 1972), we assessed three complementary components: alpha, beta, and gamma diversity. Specifically, alpha diversity was calculated as the species richness within individual soil samples. Gamma diversity was computed at the regional scale by grouping samples within each site based on their site identity, representing the total species richness within each region. Beta diversity, measuring community turnover among sites, was calculated using Bray–Curtis dissimilarity. This metric offers two key advantages: (1) resilience to data sparsity by avoiding misinterpretation of double zeros as ecological similarity, and (2) emphasis on dominant taxa, aligning with our focus on latitudinal gradients in prevalent groups. Beta diversity was assessed via multivariate dispersion analysis using the *betadisper* function in the “vegan” R package (Oksanen et al. 2026), specifically by computing the distance of each sample to the centroid of its respective regional group. To elucidate the underlying processes, we partitioned Bray–Curtis dissimilarity into two components: balanced variation in abundance (turnover) and abundance gradients (nestedness), using the “betapart” R package (Baselga and Orme 2012). This decomposition allows distinction between species replacement and richness-driven differences across sites, enabling more precise ecological inference. We calculated asymptotic diversity estimates using incidence-based (presence/absence) data to systematically evaluate sampling completeness for dominant taxa, using the “iNEXT” R package (Hsieh et al. 2016).

Latitudinal diversity patterns: Preliminary analyses fitting both linear and quadratic models to species richness–latitude

relationships at finer taxonomic resolutions (e.g., phylum or sub-phylum levels) indicated that quadratic models generally provided a better fit, as indicated by higher adjusted R^2 values and statistically significant quadratic terms ($p < 0.05$) (Figures S3–S5). Therefore, subsequent analyses of biodiversity metrics (alpha, beta, and gamma diversity) were conducted using quadratic regression models. For alpha and gamma diversity, we employed negative binomial generalized linear models (GLMs) using the *glm.nb* function in the “MASS” package to account for over-dispersed count data (Venables and Ripley 2002). To enable comparisons across groups with differing abundance ranges, we standardized diversity values using Z-score transformation within each taxonomic group. For each major category, we calculated mean standardized richness across latitudes and fitted quadratic regression models to characterize latitudinal diversity patterns. The latitude of maximum diversity for each group was determined from the peaks of these model-predicted curves. Latitudinal trends in beta diversity were analysed using beta regression model implemented in the “betareg” package in R (Cribari-Neto and Zeileis 2010). The response variable was a normalized beta diversity metric (between 0 and 1), with latitude and other covariates as predictors. To evaluate the consistency of these patterns, we subdivided the dataset into three continental regions based on geographic proximity and sample distribution, namely Africa–Europe, the Americas, and Asia–Australia, and reran the models within each continent.

Environmental drivers: To assess whether spatial clustering could influence the observed biodiversity patterns and bias the estimation of environmental effects, we first calculated global Moran's I for the species richness of each taxon using the “spdep” (Bivand and Wong 2018) and “sf” (Pebesma 2018) R packages. Statistical significance was evaluated with Monte Carlo permutation tests with 999 permutations. Spatial weights were constructed using a distance-threshold approach based on equal-area projected coordinates to avoid distortions in geographic distance, whereby samples separated by distances less than or equal to the 20th percentile of the pairwise distance distribution were considered neighbours. The resulting spatial weights matrix was row-standardized so that the weights of neighbours for each sample summed to one. Positive Moran's I values indicated spatial clustering, negative values indicated spatial dispersion, and values close to zero indicated random spatial patterns. To rigorously evaluate the environmental drivers of soil biodiversity along latitudinal gradients, we characterized the independent explanatory power of latitude, a proxy for underlying environmental gradients, on species richness using variation partitioning analysis (VPA) and hierarchical partitioning. Given that the relationship between latitude and biodiversity can be nonlinear, we performed an orthogonal polynomial expansion of latitude to extract both first-order (linear) and second-order (quadratic) terms for analysis. For beta diversity, we assessed the specific contributions of latitude via Partial Mantel tests, controlling for co-varying factors. To explore the relative influence of climate, vegetation/productivity, spatial factors, and soil properties on multitrophic soil biodiversity, we conducted a VPA. Spatial variables were derived from geographic coordinates using Moran's eigenvector maps (MEMs), generated with the *dbmem* function in the “adespatial” R package (Dray et al. 2025). To avoid overfitting, significant MEM eigenvectors were selected

through forward selection (*forward.sel*) with the adjusted R^2 of the full model as a stopping criterion. Following this, we implemented a multi-model inference using the “MuMIn” R package to identify key variables influencing soil biodiversity variations (Bartoń 2025). We initially ran a full model incorporating all predictors, treating sample locations as a random factor. This approach generated models with all possible combinations of the initial variables, which were fitted using maximum likelihood estimation and ranked by the Akaike Information Criterion (AIC). AIC values for each model were then converted to Δ AIC, representing the difference from the best-performing model. We retained all models with Δ AIC < 2 as best-fitting. For each biodiversity measure, predictor estimates across these best-fitting models were averaged using the conditional averaging approach via the *model.avg* function. For global-scale spatial prediction of biodiversity, we employed spatially explicit random forest models in ArcGIS Pro, applying them at a resolution of approximately 10 km (0.10×0.10 degrees per pixel). Regions with limited soil sample data, such as glaciers and deserts, were excluded from our spatial predictions. Prediction uncertainty was further assessed using quantile regression forest methods, implemented through the “quantregForest” R package (Meinshausen 2006).

Structural equation modelling: We applied multi-group structural equation modelling (SEM) to identify and compare the key mechanisms shaping diversity patterns across multiple taxonomic groups. The initial conceptual model was constructed based on established ecological theory, whereby latitude was hypothesized to influence alpha diversity indirectly through climate and soil properties, regional gamma diversity was included as a predictor of local alpha diversity based on the species pool hypothesis, and beta diversity was incorporated to account for spatial turnover among sites within regions (Table S1). Climatic predictors (including MAT, MAP, TSEA, and PSEA, which represent the temperature and water balance) were standardized and subjected to principal component analysis (PCA), with the first principal component (PC1) representing an integrated climatic gradient employed as an exogenous predictor in the SEM. To align the spatial resolution with the regional scale of gamma diversity, site-level alpha and beta diversity of the major taxa were aggregated into regional means. To ensure comparability across variables and improve the numerical stability of the SEM, all integrated climate and soil variables (including soil carbon, soil nitrogen, pH, and texture), as well as the three diversity components (alpha, beta, and gamma diversity), were scaled prior to model construction. Collinearity among variables was assessed using the variance inflation factor (VIF) with the “car” R package (Fox and Weisberg 2019). The optimal multi-group SEM was determined by evaluating measurement invariance across groups at four hierarchical levels: configural, metric, structural, and strict invariance. The best-fitting multi-group SEM was selected based on changes in the comparative fit index (CFI). The most constrained model with Δ CFI < 0.01 compared to the preceding model was considered to have equivalent fit and was chosen as the optimal model (Cheung and Rensvold 2002). Model fit was evaluated using chi-square statistics, degrees of freedom (df), p values, CFI, and the root mean square error of approximation (RMSEA). The final model integrates environmental factors and the beta

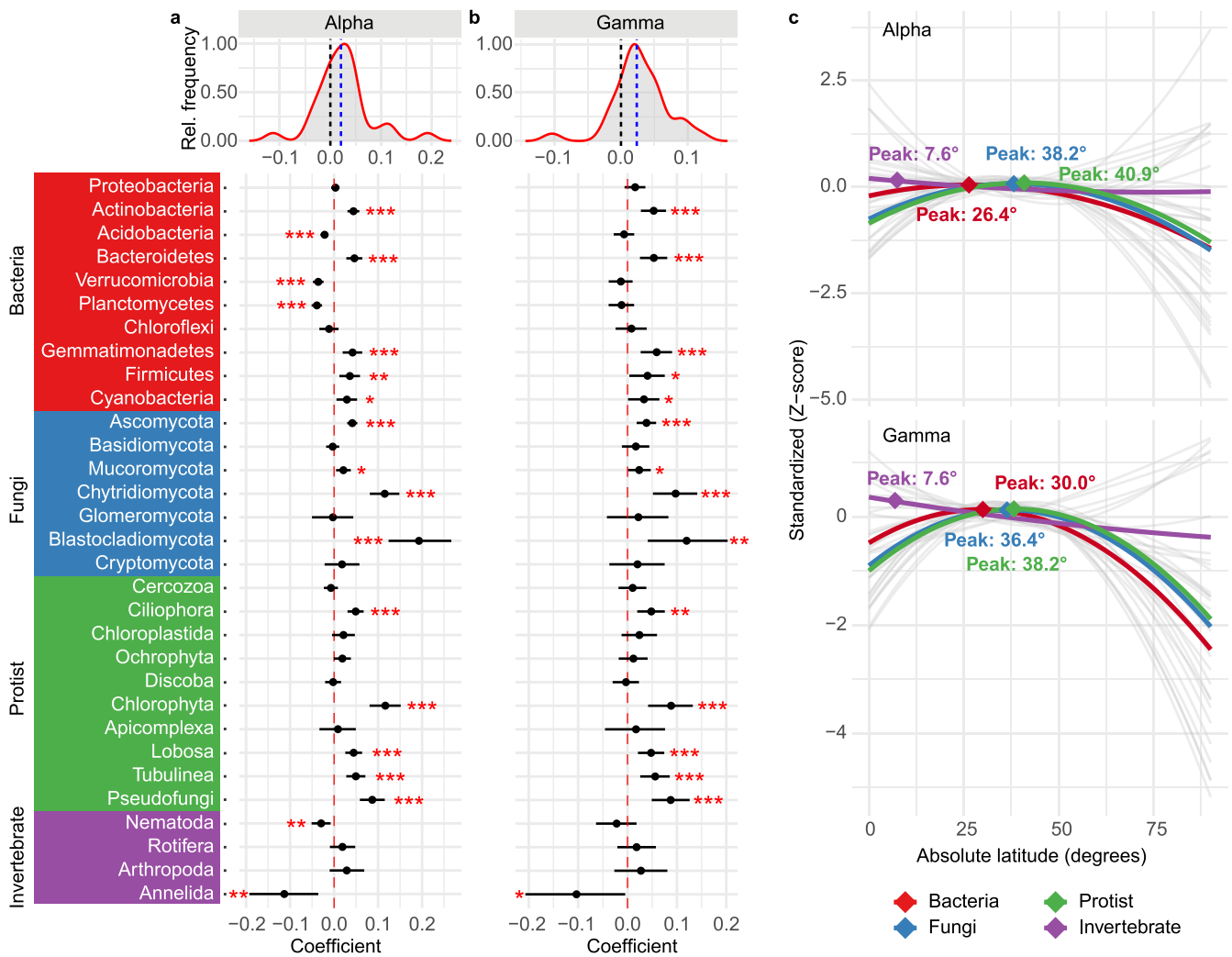


FIGURE 2 | Global latitudinal patterns of alpha (a) and gamma (b) for major microbial groups and microfauna based on global amplicon data. Estimated coefficients of the latitude–diversity relationships were derived from negative binomial generalized linear models (GLMs). Points represent the effect size (estimates) for the quadratic term of latitude, with horizontal bars indicating 95% confidence intervals. The red dashed line indicates zero effect. Asterisks denote significance levels (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). Latitudinal trends of alpha and gamma diversity fitted using GLMs (c). Grey lines represent individual taxonomic groups, while coloured lines indicate the fitted group-averaged standardized values (Z-score). Diamond symbols indicate the peak latitudes of diversity.

and gamma diversity components to explain the direct and indirect effects on alpha diversity (local species richness), with multi-group invariance testing ensuring that path structures are comparable across taxonomic groups.

3 | Results

3.1 | Latitudinal Patterns of Alpha and Gamma Diversity Across Multiple Trophic Levels

Rarefaction curves for dominant taxa reached clear asymptotic trends with increasing sampling effort, confirming that sequencing depth was sufficient and representative for robust assessments of latitudinal biodiversity patterns (Figure S6). The selected dominant taxa collectively accounted for more than 90% of total relative abundance across diverse climatic zones and ecosystems (Figures S7–S12). Comparative model testing revealed that quadratic models generally outperformed linear

models, indicating that soil biodiversity follows nonlinear latitudinal patterns (Figures S13–S14); we therefore applied negative binomial GLMs including a quadratic term to characterize latitudinal patterns in alpha and gamma diversity.

At both alpha and gamma levels, most taxa exhibited positive or nonsignificant diversity–latitude relationships, deviating markedly from the classical LDG (Figures S15–S18). For alpha diversity, only five of the 51 taxa examined showed a significant decline in richness with increasing latitude: Acidobacteria (estimate = -0.021 , $p < 0.001$), Verrucomicrobia (estimate = -0.035 , $p < 0.001$), Planctomycetes (estimate = -0.039 , $p < 0.001$), Nematoda (estimate = -0.029 , $p < 0.01$), and Annelida (estimate = -0.113 , $p < 0.01$) (Figures 2a, 3a, Tables S2–S3). These phyla varied considerably in their relative abundances (mean = 3.35%–31.58%) and detection frequencies (22.65%–100.00% of samples; Table S4). In contrast, most taxa showed positive or unimodal relationships with latitude (Figures S15–S16). For example, Chlorophyta (estimate = 0.116 , $p < 0.001$)

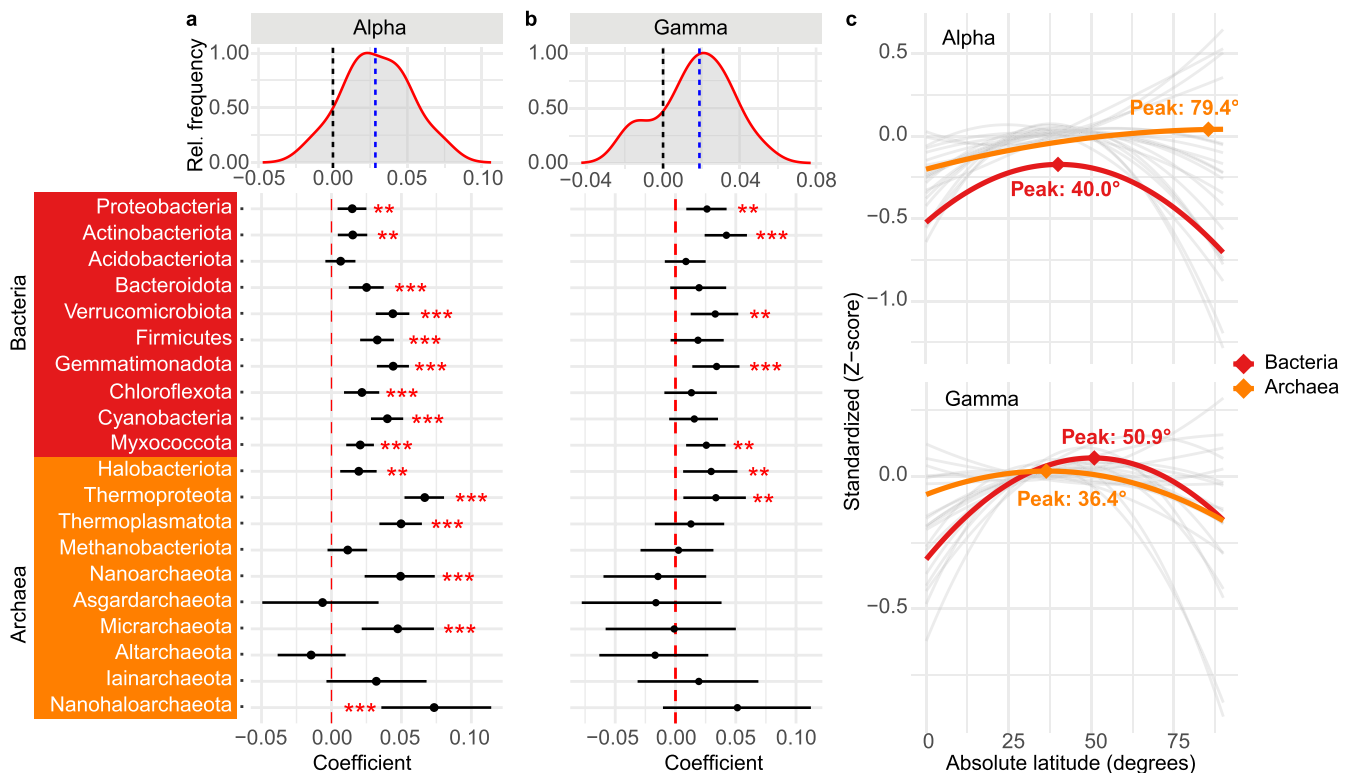


FIGURE 3 | Global latitudinal patterns of alpha (a) and gamma (b) for major microbial groups and microfauna based on global metagenomic data. Estimated coefficients of the latitude–diversity relationships were derived from negative binomial generalized linear models (GLMs). Points represent the effect size (estimates) for the quadratic term of latitude, with horizontal bars indicating 95% confidence intervals. The red dashed line indicates zero effect. Asterisks denote significance levels (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). Latitudinal trends of alpha and gamma diversity fitted using GLMs (c). Grey lines represent individual taxonomic groups, while coloured lines indicate the fitted group-averaged standardized values (Z-score). Diamond symbols indicate the peak latitudes of diversity.

and Blastocladiomycota (estimate = 0.192, $p < 0.001$) exhibited significant increases in alpha diversity with latitude in the amplicon dataset. Analysis of the metagenomic dataset revealed that none of the top 10 archaeal taxa followed the classical LDG. Instead, several major archaeal groups exhibited the opposite trend, with Thermoproteota (estimate = 0.066, $p < 0.001$) and Nanoarchaeota (estimate = 0.049, $p < 0.001$) showing significantly positive correlations with latitude. Of the remaining 46 taxa, 29 showed significantly positive correlations with latitude, while 17 showed nonsignificant relationships ($p > 0.05$; Figures 2a, 3a, Tables S2–S3). A similar trend was observed for gamma diversity: 21 of 51 taxa showed significant positive correlations with latitude (e.g., Actinobacteriota: estimate = 0.053, $p < 0.001$), while only Annelida exhibited a significant decline (estimate = -0.103, $p < 0.05$) (Figures 2b, 3b, Figures S17–S18, Tables S5–S6).

Alpha and gamma diversity exhibited broadly consistent latitudinal patterns at the global scale, though the latitude of peak diversity varied among taxonomic groups. For alpha diversity, the amplicon dataset showed that invertebrates peaked at low latitudes (7.6°), whereas bacteria, fungi, and protists peaked at approximately 26.4°, 38.2°, and 40.9°, respectively (Figure 2c). In the metagenomic dataset, the alpha diversity peak occurred at approximately 40.0° for bacteria and 79.4° for archaea (Figure 3c). To account for environmental heterogeneity and potential sampling bias, we analysed latitudinal diversity patterns separately for three continental regions, namely Africa–Europe,

the Americas, and Asia–Australia. Despite some intercontinental variation, the overall pattern persisted: most taxa displayed a hump-shaped relationship between species richness and latitude, with diversity peaking at mid-latitudes (Figures S19–S26, Tables S7–S10). The Americas and Asia–Australia exhibited the strongest latitudinal trends, whereas the Africa–Europe region showed the weakest correlation. We also observed hemispheric asymmetry: many major groups exhibited significantly higher diversity in the Northern Hemisphere than in the Southern Hemisphere, and several taxa displayed contrasting diversity–latitude relationships between hemispheres (Figures S27–S35).

As a methodological caveat, the following comparison between amplicon and metagenomic results should be interpreted qualitatively rather than as a direct numerical validation, because the two approaches differ in sample size, geographic coverage, taxonomic definition, and known detection biases. To assess the robustness of the observed patterns across sequencing approaches, we systematically compared the latitudinal diversity patterns for bacterial taxa captured by both methods. Both datasets consistently showed that most bacterial phyla deviated from the classical LDG, with diversity peaking at mid-latitudes (Figures 2, 3). However, notable quantitative differences emerged: the amplicon dataset identified the bacterial alpha diversity peak at approximately 26.4° latitude (Figure 2c), whereas the metagenomic dataset placed this peak at approximately 40.0° latitude (Figure 3c). The magnitude of latitudinal associations also differed: Acidobacteria

showed a significant negative correlation with latitude in the amplicon dataset (estimate = -0.021 , $p < 0.001$), but this relationship was weaker and nonsignificant in the metagenomic dataset (estimate = 0.006 , $p = 0.22$) (Tables S2–S3). Conversely, Proteobacteria exhibited a stronger positive relationship in the metagenomic dataset (estimate = 0.014 , $p < 0.01$) compared to the amplicon dataset (estimate = 0.003 , $p = 0.51$). For gamma diversity, similar directional consistency but quantitative divergence was observed (Tables S5–S6). Despite these methodological differences, both approaches supported the same broad biogeographic inference that most soil taxa deviate from the classical LDG. However, exact diversity values, effect sizes, and peak latitudes differed between methods and should therefore be interpreted cautiously.

3.2 | Latitudinal Variation of Beta Diversity and Its Components

Consistent with the alpha and gamma diversity patterns, beta diversity also largely deviated from the classical LDG. Across all taxonomic groups examined, only Annelida (estimate = -0.028 , $p < 0.05$) exhibited a significant negative beta diversity–latitude relationship (Figure 4a, Figures S36–S37, Tables S11–S12). In contrast, most taxa exhibited positive or nonsignificant relationships, with beta diversity generally peaking in mid- to high-latitude regions (between 35° and 50°) (Figure 4b,c). These patterns were consistent across continental regions and hemispheres, with beta diversity in the Americas being generally positively associated with latitude (Figures S38–S49). Unlike alpha diversity, beta diversity did not differ significantly between hemispheres (Figures S50–S51).

Decomposing beta diversity into species turnover and nestedness components revealed that their relative importance varied across taxa. For example, turnover dominated beta diversity in taxa such as Cyanobacteria, Nematoda, and Tubulinea, accounting for over 60% of total dissimilarity, whereas nestedness played a comparatively larger role in Annelida and Rotifera (Figures S52–S53). Both components exhibited clear latitudinal patterns. For most taxa, including Chlorophyta, Nematoda, and Asgardarchaeota, turnover and nestedness were higher around 40° latitude than in equatorial and polar regions (Figure 4b,c, Figure S54–S55). In contrast, Cryptomycota exhibited a monotonic increase in both components from low to high latitudes, whereas Ciliophora showed the opposite trend.

3.3 | Environmental Drivers of Soil Biodiversity Patterns

Moran's I analysis revealed that 39 of 51 taxa exhibited statistically significant positive spatial autocorrelation in species richness (Monte Carlo $p < 0.05$), whereas the remaining 12 taxa showed no significant spatial structure (Table S13). However, the magnitude of spatial autocorrelation was uniformly low across taxa (Moran's I range = -0.0453 to 0.0966), indicating weak spatial clustering overall. After accounting for spatial structure using Moran's eigenvector maps (MEMs) in the VPA, latitude alone explained only a small fraction of the variance in soil biodiversity (Figures S56–S57). VPA results showed that

soil properties explained 18.3%–42.6% of variation across taxa, and climate conditions explained 15.2%–38.4%, whereas geographic location accounted for only 3.1%–12.5% (Figure S58). A substantial proportion of variance remained unexplained (residuals = 28.5%–51.2%). Most taxa were positively correlated with soil C and N, but negatively correlated with TSEA, pH, and C/N ratio (Figure S59).

To identify the relative importance of individual environmental variables beyond these general trends, we applied multi-model inference, which confirmed that environmental variables were the primary predictors of soil biodiversity (mean $R^2_c = 0.66$, ranging from 0.20 to 0.96) (Figure 5, Tables S14–S15). The major determinants were soil C, soil N, MAP, MAT, and pH, though the dominant factors differed among taxa (Figure 5). Based on the summed weights of predictors, bacterial and protistan diversity were mainly influenced by MAP and soil N, fungal diversity by MAT and soil N, invertebrate diversity by soil texture and soil N, and archaeal diversity by soil N and pH.

To further explore the mechanisms shaping these patterns, we applied multi-group SEM based on an initial conceptual framework (Figure 6, Table S16). The metric model was identified as the optimal multi-group SEM after evaluating invariance at four hierarchical levels (Table S17), with excellent fit ($\chi^2/df = 0.93$, $p = 0.53$, RMSEA = 0.00, CFI = 1.00), explaining an average of 82% of the variation in diversity across taxonomic groups (Figure 6a). Latitude was only weakly associated with soil biodiversity in the SEM. In contrast, local alpha diversity was strongly and positively associated with regional gamma diversity (path coefficient = 0.87) and negatively correlated with beta diversity (path coefficient = -0.27). Path coefficients revealed clear differences in environmental responses among taxonomic groups: pH exerted significant effects on the alpha diversity of bacteria and archaea, but had weaker effects on fungi, protists, and invertebrates (Figure S60, Table S18). Total effects analysis showed that alpha diversity was mainly driven by gamma diversity and soil N (Figure 6b). Alpha diversity showed a strong, nearly linear positive relationship with gamma diversity, whereas its relationship with soil N was nonlinear; high soil N did not increase the alpha diversity of fungi, protists, or invertebrates (Figure 6c).

3.4 | Global Spatial Predictions Reveal Heterogeneous Biodiversity Patterns

Given the strong associations between soil biodiversity and environmental variables, we used spatially explicit random forest models integrating multitrophic biodiversity and environmental variables to generate a global atlas of soil biodiversity (Figure 7, Figures S61–S66, Tables S19–S22). The spatial distributions of species richness for major taxa revealed highly heterogeneous patterns that largely deviated from the LDG (Figure 7). Even within the same kingdom, contrasting patterns were evident: for example, among bacteria, Acidobacteria showed the highest richness in humid tropical regions, consistent with the classical LDG, whereas Actinobacteria peaked in arid and semi-arid subtropical zones. Similarly, fungal groups such as Ascomycota and Basidiomycota exhibited mid-latitude peaks in temperate regions, while Mucoromycota showed higher richness in tropical areas. Protist and invertebrate

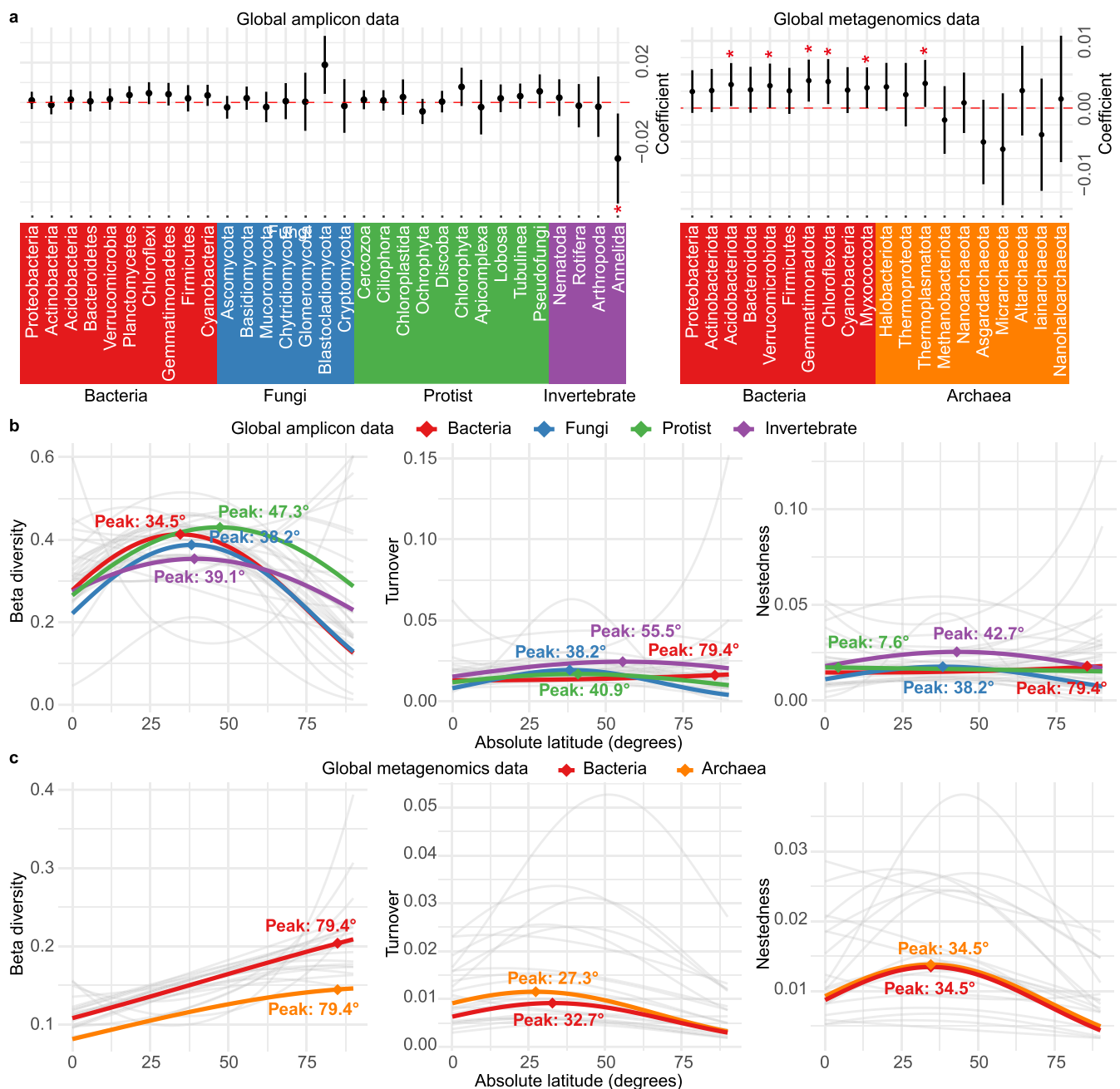


FIGURE 4 | Latitudinal patterns of beta diversity and its components (turnover and nestedness) across soil taxonomic groups. (a) Forest plots showing the estimated coefficients of latitude across taxonomic groups. Points represent mean coefficient estimates, horizontal bars indicate 95% confidence intervals, and the red dashed line marks zero effect. Asterisks denote significance levels (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). Global latitudinal patterns of beta diversity and its components (turnover and nestedness) for major microbial groups and microfauna based on (b) global amplicon and (c) metagenomic datasets. Polynomial regression models were used to fit the relationships between beta diversity and latitude. Grey lines represent individual taxonomic groups, while coloured lines indicate the fitted group-averaged trends. Diamond symbols indicate the peak latitudes of diversity.

groups also displayed divergent patterns, with Cercozoa and Nematoda peaking at mid-to-high latitudes, whereas other groups showed no clear latitudinal trend. Archaeal phyla generally exhibited sparse distributions with low richness and no discernible gradient. Overall, the spatial predictions highlighted strong heterogeneity among taxonomic groups, with most taxa showing higher diversity in temperate regions such as northern China and subtropical areas including northern and central South America, underscoring the absence of a universal LDG in soil biodiversity.

4 | Discussion

Our study provides novel and robust evidence that the global distribution of most soil organisms, including archaea, bacteria, fungi, protists, and invertebrates, largely deviates from the classical LDG, a pattern widely documented for aboveground taxa (Pianka 1966). By integrating two independent global databases spanning major biomes, we show that the expected increase in biodiversity toward the equator is generally absent or weak for most soil taxa. Instead, species richness often peaked

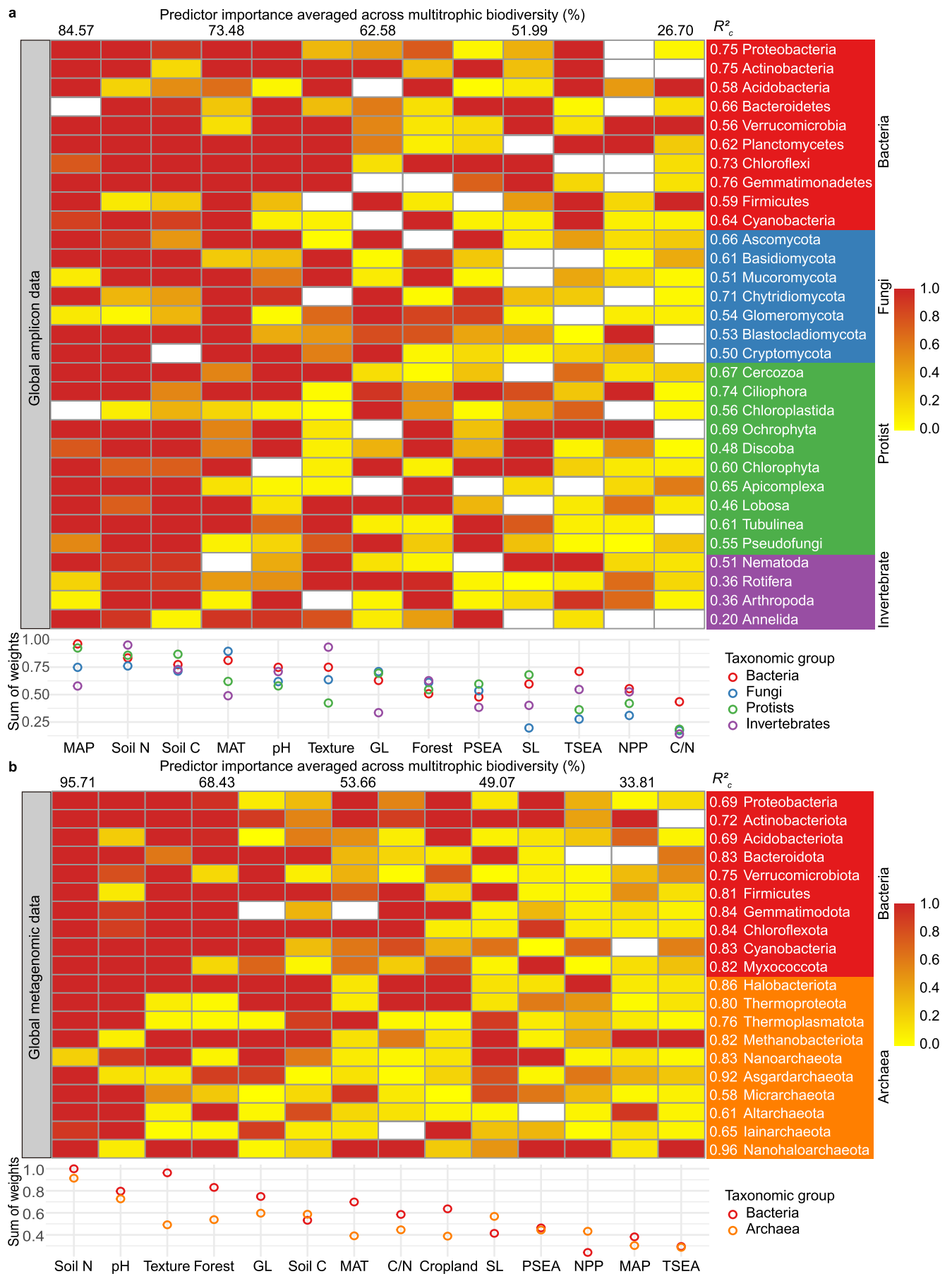


FIGURE 5 | Legend on next page.

FIGURE 5 | Relative importance of environmental predictors influencing multitrophic soil biodiversity. (a) Global amplicon data and (b) global metagenomic data were used to quantify predictor importance based on the sum of Akaike weights, proportional to the number of times a predictor appears in best-fitting models. R^2_c values indicate the proportion of variance explained. MAT, mean annual temperature; TSEA, temperature seasonality; PSEA, precipitation seasonality; MAP, mean annual precipitation; Soil C, soil carbon; Soil N, soil nitrogen; C/N, carbon: nitrogen ratio; Texture, clay + silt %; GL, grassland; SL, shrubland; NPP, net primary productivity.

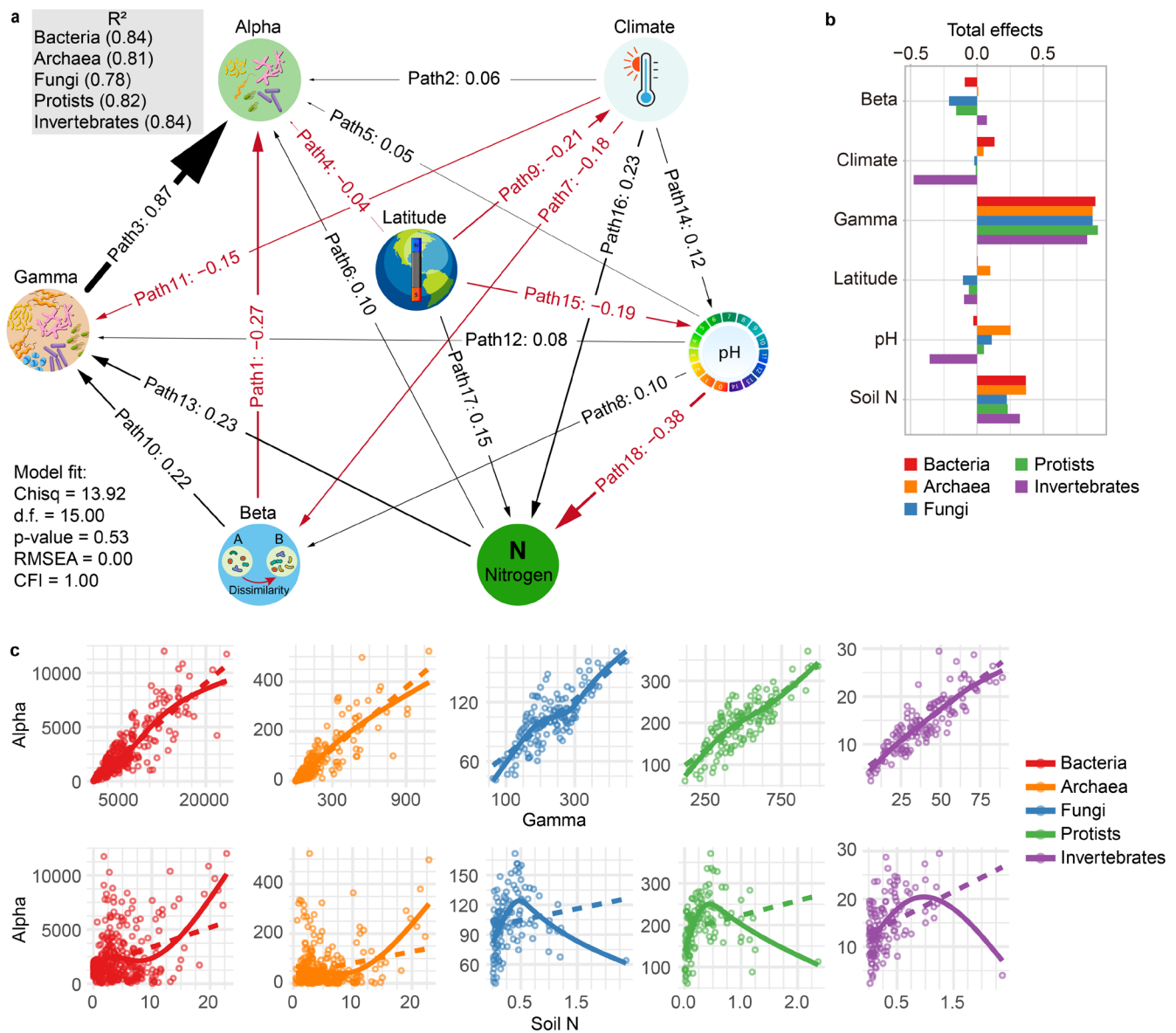


FIGURE 6 | (a) Multi-group structural equation models revealed the key determinants of soil biodiversity across different taxonomic groups. (b) Black and red arrows represent positive and negative relationships, respectively. Numbers on the paths indicate the average standardized path coefficients of all taxa, and arrow thickness corresponds to the magnitude of the coefficients. Total standardized effects of major predictors on alpha diversity. (c) Bars represent total effect values for each taxonomic group, with positive and negative values indicating promoting or inhibitory effects, respectively. Relationships between alpha diversity and gamma diversity and soil N. Each panel shows a specific group with both linear (dashed lines) and nonlinear (solid lines; fitted using generalized additive models [GAMs]) fits.

in temperate regions (around 39° latitude, Figures 2c, 3c), and beta diversity reached its maximum at mid- to high latitudes (Figure 4).

As the first global multitrophic assessment spanning major soil domains, our findings reveal fundamental differences between

aboveground and belowground biogeographic patterns. These results align with previous studies reporting inconsistent or weak LDG patterns in soil bacteria (Fierer and Jackson 2006) and Collembola (Potapov et al. 2024), while contrasting with reports of classical LDG in fungi (Tedersoo et al. 2014). Notably, Annelida was the only group conforming to the classical LDG

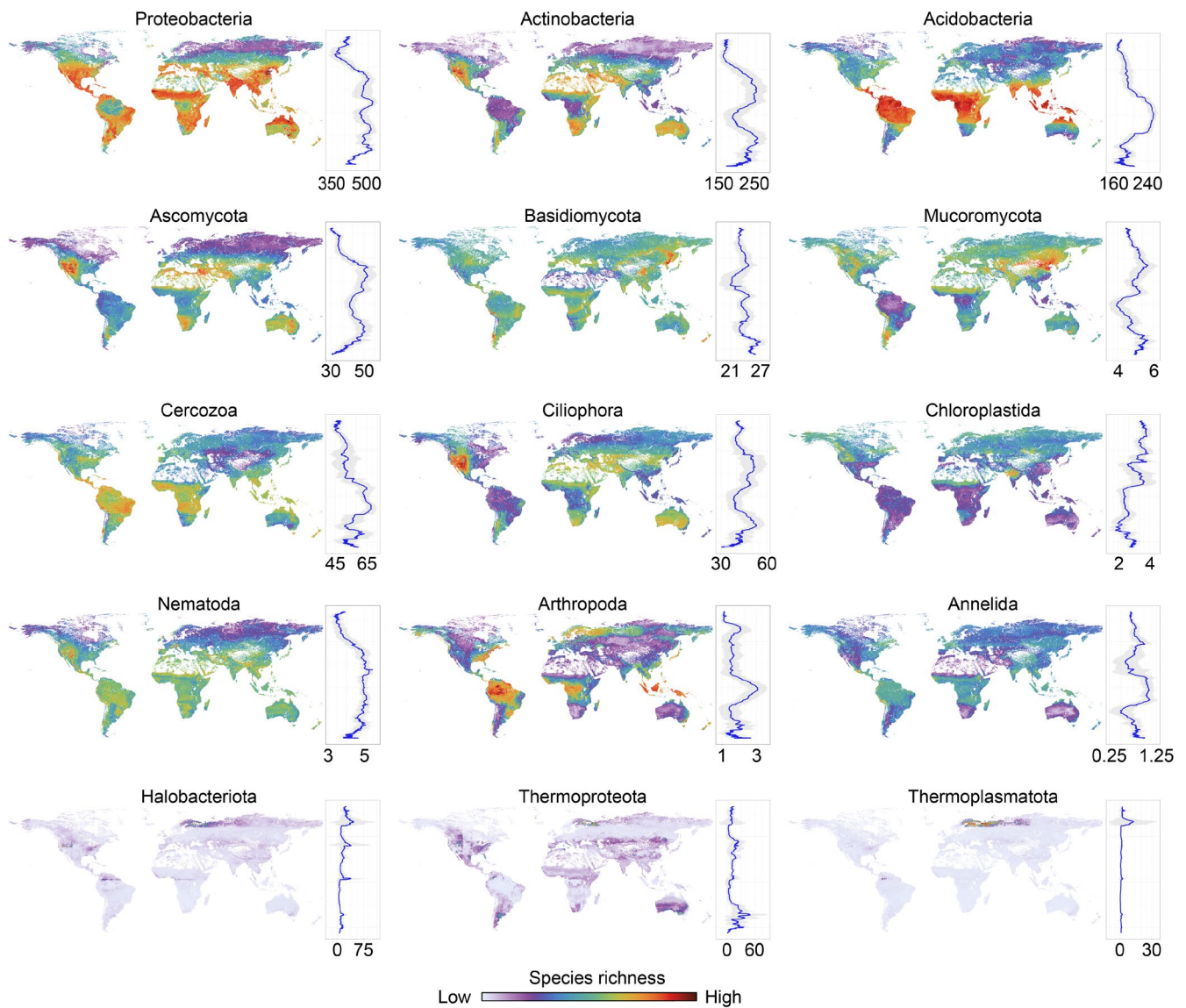


FIGURE 7 | Global predicted distributions of selected dominant microbial and faunal richness, generated using spatially explicit random forest models at $0.10^{\circ} \times 0.10^{\circ}$ spatial resolution (~ 10 km). Grey areas indicate regions excluded from the analysis because of limited soil sample data, such as glaciers and deserts. The side panels show the latitudinal patterns of predicted richness for each taxon.

across all three diversity dimensions (Figures 2–4), likely reflecting its limited dispersal capacity (meters per year) and lack of passive long-distance dispersal, making their distributions more strongly constrained by historical biogeographic processes (Hendrix et al. 2008; Marinissen and van den Bosch 1992). In contrast, archaea represented a particularly striking deviation, with none of the top 10 archaeal taxa showing declining diversity toward higher latitudes (Figure 3), consistent with the adaptation of several archaeal lineages, such as ammonia-oxidizing Thaumarchaeota, to oligotrophic and cold environments (Alves et al. 2018; Prosser and Nicol 2012). Together, these results underscore that latitudinal diversity patterns in soils are fundamentally taxon-dependent. These patterns challenge conservation strategies focused exclusively on tropical regions (Thomson et al. 2018), given that soil biodiversity hotspots are concentrated in temperate regions (Figure 7).

Our analyses consistently show that local edaphic factors (soil C, N, and pH) and broader-scale climate variables (MAT, MAP)

explain far more biodiversity variation than latitude alone (mean $R^2_c = 0.66$; Figure 5), with the relative importance of individual drivers varying among taxa. This supports previous findings that soil biodiversity is more closely linked to local soil properties (Delgado-Baquerizo et al. 2018; Fierer and Jackson 2006; Mod et al. 2021) and climate (Tian et al. 2017) than to latitude per se. The temperate peak in diversity may thus arise from optimal combinations of temperature, moisture, and soil nutrient availability, which create favourable niches for diverse soil taxa (Biryol et al. 2024). Moreover, temperate areas may serve as biogeographical transition zones, harbouring taxa adapted to both tropical and polar conditions (Bernardo-Madrid et al. 2025).

Beta diversity, largely dominated by species turnover ($> 60\%$) rather than nestedness in most taxa (such as Chlorophyta, Nematoda, and Asgardarchaeota), peaked at mid- to high latitudes rather than in tropical regions (Figure 4). This pattern is consistent with findings from eastern China (Zhang et al. 2020) and suggests that community differentiation is strongest where

environmental heterogeneity is greatest. In temperate regions, stronger seasonality in temperature, moisture, and nutrient availability, combined with topographic complexity and diverse vegetation types, likely creates a mosaic of microhabitats that promotes species replacement and elevates regional biodiversity, as supported by the observed highest gamma diversity in these areas. In contrast, tropical soils may exhibit lower beta diversity due to more stable environmental conditions that reduce niche partitioning (Bachar et al. 2010). Our findings highlight the scale-dependence of soil biodiversity gradients: alpha diversity tends to peak where local resources are favourable, whereas beta diversity is highest where environmental variability promotes community turnover.

Our multi-group SEM analyses revealed that alpha diversity was jointly constrained by regional gamma diversity (total effects=0.87) and soil N (total effects=0.30) (Figure 6), highlighting a dual control mechanism in which regional processes determine the potential species pool, while local edaphic filters govern which taxa persist. Several mechanisms may help explain why soil biodiversity deviates from the classical LDG. First, stronger environmental heterogeneity in temperate zones, driven by climatic seasonality and topographic complexity, may support higher gamma and beta diversity, consistent with the environmental heterogeneity hypothesis (Stein et al. 2014). Second, soil N showed taxon-specific and nonlinear relationships with alpha diversity, indicating that nutrient limitation can regulate local coexistence differently across trophic groups (Marshall 2007). In fungi, protists, and invertebrates, diversity increased up to a threshold and then declined at high N levels, whereas bacteria and archaea showed U-shaped responses (Figure 6c), providing only limited and taxon-dependent support for the intermediate productivity hypothesis. Third, the high dispersal capacity of many soil microbes may weaken latitudinal gradients by promoting widespread colonization across latitudes (Bahram et al. 2018), whereas dispersal-limited taxa such as earthworms may remain more strongly structured by historical biogeography. This may explain why several taxa followed the LDG in alpha diversity, but only Annelida conformed to the classical LDG in gamma diversity (Figures 2, 3). We acknowledge that plant community characteristics, including plant diversity, vegetation type, and litter quality, also influence soil biodiversity but were not fully captured in our analyses. Although NPP was included as a proxy for plant productivity, the relatively low conditional R^2 values for some responses (ranging from 0.20 to 0.96; Figure 5) likely reflect, in part, the influence of these unmeasured vegetation-related factors.

Our dual-method approach yielded consistent broad conclusions but also highlighted important methodological differences that limit direct quantitative comparability. Amplicon sequencing provided high sensitivity but was constrained by primer biases (McNichol et al. 2021), whereas metagenomics offers broader taxonomic coverage, although low-abundance organisms may still be underrepresented (Liu et al. 2025). The bacterial diversity peak differed between approaches (Figures 2c, 3c), likely reflecting primer bias (e.g., underrepresentation of Verrucomicrobia by 515F/806R) (Parada et al. 2016), unequal sample size (4856 vs. 615), differences in high-latitude coverage, and differences in taxonomic resolution between zOTUs and ANI-based metagenomic clusters. These discrepancies underscore the need for

methodological cross-validation, as many foundational studies have relied exclusively on amplicon sequencing (Bahram et al. 2018; Fierer and Jackson 2006). Nevertheless, because both approaches converged on the same nonclassical LDG pattern, our principal qualitative conclusion is robust to method choice, even though the precise magnitude and location of diversity peaks remain method-dependent.

Despite our extensive dataset (>5,000 samples) spanning major biomes worldwide, sampling gaps persist in tropical Africa and South America, and the overall dataset is biased toward the Northern Hemisphere (Figure 1). Although environmental factors explain substantial variation, other ecological and evolutionary processes, including ecological drift, dispersal, and speciation, likely also contribute to observed patterns. Future research should integrate better coverage of undersampled regions, more detailed vegetation data, and explicit comparisons across both dominant and rare taxa. Such efforts will be essential for refining a general theory of soil biogeography and for understanding how soil biodiversity underpins ecosystem functioning and resilience under global change (Crowther et al. 2024).

5 | Conclusion

Our findings challenge the universality of the classical LDG by demonstrating that this pattern does not consistently apply to soil microbes and fauna—the largest component of Earth's biodiversity. By explicitly examining multitrophic soil biodiversity, our study provides novel insights into the latitudinal distribution of soil biodiversity and reveals that soil biodiversity patterns emerge from the interplay among regional species pools, local nutrient levels, and spatial environmental heterogeneity. This study advances a revised framework for global soil biodiversity biogeography, one that emphasizes environmental filtering, resource availability, and taxon-specific ecological strategies as the primary determinants of diversity across Earth's soils. These insights are essential for guiding effective conservation strategies and for understanding the role of soil organisms in maintaining ecosystem stability and resilience under accelerating global change.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data and scripts have been deposited to the Figshare repository (<https://doi.org/10.6084/m9.figshare.28022270>).

References

- Alves, R. J. E., B. Q. Minh, T. Ulrich, A. von Haeseler, and C. Schleper. 2018. "Unifying the Global Phylogeny and Environmental Distribution of Ammonia-Oxidising Archaea Based on *amoA* Genes." *Nature Communications* 9: 1517.
- Amaral-Zettler, L. A., E. A. McCliment, H. W. Ducklow, and S. M. Huse. 2009. "A Method for Studying Protistan Diversity Using Massively Parallel Sequencing of V9 Hypervariable Regions of Small-Subunit Ribosomal RNA Genes." *PLoS One* 4: e6372.
- Anthony, M. A., S. F. Bender, and M. G. van der Heijden. 2023. "Enumerating Soil Biodiversity." *Proceedings of the National Academy of Sciences of the United States of America* 120: e2304663120.
- Averill, C., M. A. Anthony, P. Baldrian, et al. 2022. "Defending Earth's Terrestrial Microbiome." *Nature Microbiology* 7: 1717–1725.
- Bachar, A., A. Al-Ashhab, M. I. Soares, et al. 2010. "Soil Microbial Abundance and Diversity Along a Low Precipitation Gradient." *Microbial Ecology* 60: 453–461.
- Bahram, M., F. Hildebrand, S. K. Forslund, et al. 2018. "Structure and Function of the Global Topsoil Microbiome." *Nature* 560: 233–237.
- Banerjee, S., and M. G. A. van der Heijden. 2023. "Soil Microbiomes and One Health." *Nature Reviews Microbiology* 21: 6–20.
- Bartoń, K. 2025. "MuMin: Multi-Model Inference (Version R package version 1.48.11)." <https://CRAN.R-project.org/package=MuMin>.
- Baselga, A., and C. D. L. Orme. 2012. "Betapart: An R Package for the Study of Beta Diversity." *Methods in Ecology and Evolution* 3: 808–812.
- Bernardo-Madrid, R., M. Gonzalez-Suarez, M. Rosvall, et al. 2025. "A General Rule on the Organization of Biodiversity in Earth's Biogeographical Regions." *Nature Ecology & Evolution* 9: 1193–1204.
- Biryol, C., A. Aupic-Samain, C. Lecareux, T. Gauquelin, V. Baldy, and M. Santonja. 2024. "Interactive Effects of Soil Moisture, Air Temperature and Litter Nutrient Diversity on Soil Microbial Communities and Population." *Oikos* 2024: e10345.
- Bivand, R. S., and D. W. S. Wong. 2018. "Comparing Implementations of Global and Local Indicators of Spatial Association." *TEST* 27: 716–748.
- Bolger, A. M., M. Lohse, and B. Usadel. 2014. "Trimmomatic: A Flexible Trimmer for Illumina Sequence Data." *Bioinformatics* 30: 2114–2120.
- Caporaso, J. G., J. Kuczynski, J. Stombaugh, et al. 2010. "QIIME Allows Analysis of High-Throughput Community Sequencing Data." *Nature Methods* 7: 335–336.
- Cheung, G. W., and R. B. Rensvold. 2002. "Evaluating Goodness-Of-Fit Indexes for Testing Measurement Invariance." *Structural Equation Modeling: A Multidisciplinary Journal* 9: 233–255.
- Condamine, F. L., F. A. Sperling, N. Wahlberg, J. Y. Rasplus, and G. J. Kergoat. 2012. "What Causes Latitudinal Gradients in Species Diversity? Evolutionary Processes and Ecological Constraints on Swallowtail Biodiversity." *Ecology Letters* 15: 267–277.
- Cribari-Neto, F., and A. Zeileis. 2010. "Beta Regression in R." *Journal of Statistical Software* 34: 1–24.
- Crowther, T. W., R. Rappuoli, C. Corinaldesi, et al. 2024. "Scientists' Call to Action: Microbes, Planetary Health, and the Sustainable Development Goals." *Cell* 187: 5195–5216.
- Crowther, T. W., J. van den Hoogen, J. Wan, et al. 2019. "The Global Soil Community and Its Influence on Biogeochemistry." *Science* 365: eaav0550.
- Delgado-Baquerizo, M., R. D. Bardgett, P. M. Vitousek, et al. 2019. "Changes in Belowground Biodiversity During Ecosystem Development." *Proceedings of the National Academy of Sciences of the United States of America* 116: 6891–6896.
- Delgado-Baquerizo, M., A. M. Oliverio, T. E. Brewer, et al. 2018. "A Global Atlas of the Dominant Bacteria Found in Soil." *Science* 359: 320–325.
- Dray, S., D. Bauman, G. Blanchet, et al. 2025. "adespatial: Multivariate Multiscale Spatial Analysis (Version R package version 0.3–28)." <https://CRAN.R-project.org/package=adespatial>.
- Duchêne, D. A., and M. Cardillo. 2015. "Phylogenetic Patterns in the Geographic Distributions of Birds Support the Tropical Conservatism Hypothesis." *Global Ecology and Biogeography* 24: 1261–1268.
- Edgar, R. C. 2010. "Search and Clustering Orders of Magnitude Faster Than BLAST." *Bioinformatics* 26: 2460–2461.
- Edgar, R. C. 2016. "UNOISE2: Improved Error-Correction for Illumina 16S and ITS Amplicon Sequencing." *bioRxiv*. <https://doi.org/10.1101/081257>.
- Eldridge, D. J., E. Guirado, P. B. Reich, et al. 2023. "The Global Contribution of Soil Mosses to Ecosystem Services." *Nature Geoscience* 16: 430–438.
- Fierer, N., and R. B. Jackson. 2006. "The Diversity and Biogeography of Soil Bacterial Communities." *Proceedings of the National Academy of Sciences of the United States of America* 103: 626–631.
- Fox, J., and S. Weisberg. 2019. *An R Companion to Applied Regression*. Third edition ed. Sage.
- Guerra, C. A., M. Berdugo, D. J. Eldridge, et al. 2022. "Global Hotspots for Soil Nature Conservation." *Nature* 610: 693–698.
- Guerra, C. A., A. Heintz-Buschart, J. Sikorski, et al. 2020. "Blind Spots in Global Soil Biodiversity and Ecosystem Function Research." *Nature Communications* 11: 3870.
- Hendrix, P. F., M. Callahan, J. M. Drake, et al. 2008. "Pandora's Box Contained Bait: The Global Problem of Introduced Earthworms." *Annual Review of Ecology, Evolution, and Systematics* 39: 593–613.
- Hillebrand, H. 2004. "On the Generality of the Latitudinal Diversity Gradient." *American Naturalist* 163: 192–211.
- Hsieh, T. C., K. H. Ma, and A. Chao. 2016. "iNEXT: An R Package for Rarefaction and Extrapolation of Species Diversity (Hill Numbers)." *Methods in Ecology and Evolution* 7: 1451–1456.
- Hyde, K. D., S. Fryar, Q. Tian, A. H. Bahkali, and J. Xu. 2016. "Lignicolous Freshwater Fungi Along a North–South Latitudinal Gradient in the Asian/Australian Region; Can We Predict the Impact of Global Warming on Biodiversity and Function?" *Fungal Ecology* 19: 190–200.
- Ji, M. K., W. D. Kong, H. Z. Jia, et al. 2022. "Polar Soils Exhibit Distinct Patterns in Microbial Diversity and Dominant Phylotypes." *Soil Biology and Biochemistry* 166: 108550.
- Jun, C., Y. Ban, and S. Li. 2014. "China: Open Access to Earth Land-Cover Map." *Nature* 514, no. 7523: 434.
- Kettler, T. A., J. W. Doran, and T. L. Gilbert. 2001. "Simplified Method for Soil Particle-Size Determination to Accompany Soil-Quality Analyses." *Soil Science Society of America Journal* 65: 849–852.
- Li, D., C. M. Liu, R. Luo, K. Sadakane, and T. W. Lam. 2015. "MEGAHIT: An Ultra-Fast Single-Node Solution for Large and Complex Metagenomics Assembly via Succinct de Bruijn Graph." *Bioinformatics* 31: 1674–1676.

- Liu, S., J. S. Rodriguez, V. Munteanu, et al. 2025. "Analysis of Metagenomic Data." *Nature Reviews Methods Primers* 5: 5.
- Ma, B., C. Lu, Y. Wang, et al. 2023. "A Genomic Catalogue of Soil Microbiomes Boosts Mining of Biodiversity and Genetic Resources." *Nature Communications* 14: 7318.
- Marcot, J. D., D. L. Fox, and S. R. Niebuhr. 2016. "Late Cenozoic Onset of the Latitudinal Diversity Gradient of North American Mammals." *Proceedings of the National Academy of Sciences of the United States of America* 113: 7189–7194.
- Marinissen, J. C., and F. van den Bosch. 1992. "Colonization of New Habitats by Earthworms." *Oecologia* 91: 371–376.
- Marshall, C. R. 2007. "Explaining Latitudinal Diversity Gradients." *Science* 317: 451–453.
- Martiny, J. B., B. J. Bohannan, J. H. Brown, et al. 2006. "Microbial Biogeography: Putting Microorganisms on the Map." *Nature Reviews Microbiology* 4: 102–112.
- McNichol, J., P. M. Berube, S. J. Biller, and J. A. Fuhrman. 2021. "Evaluating and Improving Small Subunit rRNA PCR Primer Coverage for Bacteria, Archaea, and Eukaryotes Using Metagenomes From Global Ocean Surveys." *mSystems* 6: e0056521.
- Meinshausen, N. 2006. "Quantile Regression Forests." *Journal of Machine Learning Research* 7: 983–999.
- Mittelbach, G. G., and B. J. McGill. 2019. *Community Ecology*. Second edition ed. Oxford University Press.
- Mittelbach, G. G., D. W. Schemske, H. V. Cornell, et al. 2007. "Evolution and the Latitudinal Diversity Gradient: Speciation, Extinction and Biogeography." *Ecology Letters* 10: 315–331.
- Mod, H. K., A. Buri, E. Yashiro, et al. 2021. "Predicting Spatial Patterns of Soil Bacteria Under Current and Future Environmental Conditions." *ISME Journal* 15: 2547–2560.
- Nayfach, S., D. Paez-Espino, L. Call, et al. 2021. "Metagenomic Compendium of 189,680 DNA Viruses From the Human Gut Microbiome." *Nature Microbiology* 6: 960–970.
- Oksanen, J., G. Simpson, F. Blanchet, et al. 2026. "vegan: Community Ecology Package (Version R package version 2.7-3)". <https://vegandevs.github.io/vegan/>.
- Oliverio, A. M., S. Geisen, M. Delgado-Baquerizo, F. T. Maestre, B. L. Turner, and N. Fierer. 2020. "The Global-Scale Distributions of Soil Protists and Their Contributions to Belowground Systems." *Science Advances* 6: eaax8787.
- Orgiazzi, A., R. D. Bardgett, E. Barrios, et al. 2016. *Global Soil Biodiversity Atlas*. European Commission: Publications Office of the European Union.
- Ounit, R., and S. Lonardi. 2016. "Higher Classification Sensitivity of Short Metagenomic Reads With CLARK-S." *Bioinformatics* 32: 3823–3825.
- Parada, A. E., D. M. Needham, and J. A. Fuhrman. 2016. "Every Base Matters: Assessing Small Subunit rRNA Primers for Marine Microbiomes With Mock Communities, Time Series and Global Field Samples." *Environmental Microbiology* 18: 1403–1414.
- Pebesma, E. 2018. "Simple Features for R: Standardized Support for Spatial Vector Data." *R Journal* 10: 439–446.
- Phillips, H. R. P., C. A. Guerra, M. L. C. Bartz, et al. 2019. "Global Distribution of Earthworm Diversity." *Science* 366: 480–485.
- Pianka, E. R. 1966. "Latitudinal Gradients in Species Diversity – a Review of Concepts." *American Naturalist* 100: 33–46.
- Potapov, A. M., T. W. Chen, A. V. Striuchkova, et al. 2024. "Global Fine-Resolution Data on Springtail Abundance and Community Structure." *Scientific Data* 11: 22.
- Prosser, J. I., and G. W. Nicol. 2012. "Archaeal and Bacterial Ammonia-Oxidisers in Soil: The Quest for Niche Specialisation and Differentiation." *Trends in Microbiology* 20: 523–531.
- Ramirez, K. S., J. W. Leff, A. Barberan, et al. 2014. "Biogeographic Patterns in Below-Ground Diversity in New York City's Central Park Are Similar to Those Observed Globally." *Proceedings of the Royal Society B: Biological Sciences* 281: 20141988.
- Shade, A., R. R. Dunn, S. A. Blowes, et al. 2018. "Macroecology to Unite All Life, Large and Small." *Trends in Ecology & Evolution* 33: 731–744.
- Stein, A., K. Gerstner, and H. Kreft. 2014. "Environmental Heterogeneity as a Universal Driver of Species Richness Across Taxa, Biomes and Spatial Scales." *Ecology Letters* 17: 866–880.
- Stoeck, T., D. Bass, M. Nebel, et al. 2010. "Multiple Marker Parallel Tag Environmental DNA Sequencing Reveals a Highly Complex Eukaryotic Community in Marine Anoxic Water." *Molecular Ecology* 19: 21–31.
- Tedersoo, L., M. Bahram, S. Polme, et al. 2014. "Fungal Biogeography. Global Diversity and Geography of Soil Fungi." *Science* 346: 1256688.
- Tedersoo, L., M. Bahram, M. Toots, et al. 2012. "Towards Global Patterns in the Diversity and Community Structure of Ectomycorrhizal Fungi." *Molecular Ecology* 21: 4160–4170.
- Thomson, S. A., R. L. Pyle, S. T. Ah Yong, et al. 2018. "Taxonomy Based on Science Is Necessary for Global Conservation." *PLoS Biology* 16: e2005075.
- Tian, J., N. He, L. Hale, et al. 2017. "Soil Organic Matter Availability and Climate Drive Latitudinal Patterns in Bacterial Diversity From Tropical to Cold Temperate Forests." *Functional Ecology* 32: 61–70.
- van den Hoogen, J., S. Geisen, D. Routh, et al. 2019. "Soil Nematode Abundance and Functional Group Composition at a Global Scale." *Nature* 572: 194–198.
- Venables, W. N., and B. D. Ripley. 2002. *Modern Applied Statistics With S*. Fourth edition ed. Springer.
- Wang, X., J. D. Van Nostrand, Y. Deng, et al. 2015. "Scale-Dependent Effects of Climate and Geographic Distance on Bacterial Diversity Patterns Across Northern China's Grasslands." *FEMS Microbiology Ecology* 91: fiv133.
- Whittaker, R. H. 1972. "Evolution and Measurement of Species Diversity." *Taxon* 21: 213–251.
- Wiens, J. J., and M. J. Donoghue. 2004. "Historical Biogeography, Ecology and Species Richness." *Trends in Ecology & Evolution* 19: 639–644.
- Willig, M. R., D. M. Kaufman, and R. D. Stevens. 2003. "Latitudinal Gradients of Biodiversity: Pattern, Process, Scale, and Synthesis." *Annual Review of Ecology, Evolution, and Systematics* 34: 273–309.
- Zhang, X., S. Liu, J. Wang, et al. 2020. "Local Community Assembly Mechanisms Shape Soil Bacterial Beta Diversity Patterns Along a Latitudinal Gradient." *Nature Communications* 11: 5428.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Changes in soil properties with latitude in the two hemispheres. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. MAT, mean annual temperature; TSEA, temperature seasonality; PSEA, precipitation seasonality; MAP, mean annual precipitation; Soil C, soil carbon; Soil N, soil nitrogen; C/N, Soil carbon: nitrogen ratio; Texture, clay + silt %. The data originated from global amplicon data. **Figure S2:** Changes in soil properties with latitude in the two hemispheres. Data were fitted using both linear and quadratic models. Linear regressions were shown with

dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. MAT, mean annual temperature; TSEA, temperature seasonality; PSEA, precipitation seasonality; MAP, mean annual precipitation; Soil C, soil carbon; Soil N, soil nitrogen; C/N, Soil carbon: nitrogen ratio; Texture, sand %. The data originated from global metagenomic data. **Figure S3:** Changes in soil biodiversity with latitude in the two hemispheres. Variations in the microbial richness with latitude for the major microbial groups of bacteria, fungi, protists, and invertebrates. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. The data originated from global amplicon data. **Figure S4:** Changes in soil biodiversity with latitude in the two hemispheres. Variations in the microbial richness with latitude for the major microbial groups of archaea. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. The data originated from global metagenomic data. **Figure S5:** Changes in soil biodiversity with latitude in the two hemispheres. Variations in the microbial richness with latitude for the major microbial groups of bacteria. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. The data originated from global amplicon data. **Figure S6:** Rarefaction curves (Hill number $q = 0$) for the species richness of the major taxa across sampling sites. Analyses were performed based on incidence-based occurrence data, using the iNEXT R package. **Figure S7:** Changes in bacterial communities with latitude in the two hemispheres. (a) The relative abundances of the major bacterial populations among different climatic zones. (b) The changes in the relative abundance of the major microbial groups with latitude. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. The data originated from global amplicon data. **Figure S8:** Changes in fungal communities with latitude in the two hemispheres. (a) The relative abundances of the major fungal populations among different climatic zones. (b) The changes in the relative abundance of the major microbial groups with latitude. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. The data originated from global amplicon data. **Figure S9:** Changes in soil protist communities with latitude in the two hemispheres. (a) The relative abundances of the major protist populations among different climatic zones. (b) The changes in the relative abundance of the major microbial groups with latitude. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. The data originated from global amplicon data. **Figure S10:** Changes in soil invertebrate communities with latitude in the two hemispheres. (a) The relative abundances of the major invertebrate populations among different climatic zones. (b) The changes in the relative abundance of the major microbial groups with latitude. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. The data originated from global amplicon data. **Figure S11:** The relative abundances of the major microbial populations among different land covers for (a) archaea and (b) bacterial communities. The data originated from global metagenomic data. **Figure S12:** The changes in the relative abundance of the

major microbial groups with latitude. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. The data originated from global metagenomic data. **Figure S13:** Changes in soil biodiversity with latitude in the two hemispheres. Variations in the microbial richness with latitude for the major microbial groups of bacteria, fungi, protists, and invertebrates. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. The data originated from global amplicon data. **Figure S14:** Changes in soil biodiversity with latitude in the two hemispheres. Variations in the microbial richness with latitude for the major microbial groups of archaea. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. The data originated from global metagenomic data. **Figure S15:** Relationship between species richness and absolute latitude. Scatter points represent observed richness values for different taxa. Curves show fitted relationships from negative binomial GLMs including a quadratic term. The data originated from global amplicon data. **Figure S16:** Relationship between species richness and absolute latitude. Scatter points represent observed richness values for different taxa. Curves show fitted relationships from negative binomial GLMs including a quadratic term. The data originated from global metagenomic data. **Figure S17:** Relationship between gamma diversity and absolute latitude. Scatter points represent gamma diversity values for different taxa. Curves show fitted relationships from negative binomial GLMs including a quadratic term. The data originated from global amplicon data. **Figure S18:** Relationship between gamma diversity and absolute latitude. Scatter points represent gamma diversity values for different taxa. Curves show fitted relationships from negative binomial generalized linear models (GLMs) including a quadratic term. The data originated from global metagenomic data. **Figure S19:** Relationship between species richness and absolute latitude across continental regions. Curves show fitted relationships from negative binomial GLMs including a quadratic term. The data originated from global amplicon data. **Figure S20:** Estimated coefficients of the latitude–alpha diversity relationships were derived from negative binomial generalized linear models (GLMs). Points represent the effect size (estimates) for the quadratic term of latitude, with horizontal bars indicating 95% confidence intervals. The red dashed line indicates zero effect. Asterisks denote significance levels ($*p < 0.05$; $**p < 0.01$; $***p < 0.001$). The data originated from global amplicon data. **Figure S21:** Relationship between species richness and absolute latitude across continental regions. Curves show fitted relationships from negative binomial generalized linear models (GLMs) including a quadratic term. The data originated from global metagenomic data. **Figure S22:** Estimated coefficients of the latitude–alpha diversity relationships were derived from negative binomial generalized linear models (GLMs). Points represent the effect size (estimates) for the quadratic term of latitude, with horizontal bars indicating 95% confidence intervals. The red dashed line indicates zero effect. Asterisks denote significance levels ($*p < 0.05$; $**p < 0.01$; $***p < 0.001$). The data originated from global metagenomic data. **Figure S23:** Relationship between gamma diversity and absolute latitude across continental regions. Curves show fitted relationships from negative binomial GLMs including a quadratic term. The data originated from global amplicon data. **Figure S24:** Estimated coefficients of the latitude–gamma diversity relationships were derived from negative binomial generalized linear models (GLMs). Points represent the effect size (estimates) for the quadratic term of latitude, with horizontal bars indicating 95% confidence intervals. The red dashed line indicates zero effect. Asterisks denote significance levels ($*p < 0.05$; $**p < 0.01$; $***p < 0.001$). The data originated from global amplicon data. **Figure S25:** Relationship between gamma diversity and absolute latitude across continental regions. Curves show fitted relationships from negative binomial generalized linear models (GLMs) including a

quadratic term. The data originated from global metagenomic data. **Figure S26:** Estimated coefficients of the latitude–gamma diversity relationships were derived from negative binomial generalized linear models (GLMs). Points represent the effect size (estimates) for the quadratic term of latitude, with horizontal bars indicating 95% confidence intervals. The red dashed line indicates zero effect. Asterisks denote significance levels (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). The data originated from global metagenomic data. **Figure S27:** Mean values for soil biodiversity in two hemispheres. The difference in the richness of major microbial groups between the North and South hemispheres. The significant difference was examined using Wilcoxon rank sum tests. The data originated from global amplicon data. **Figure S28:** Mean values for soil biodiversity in two hemispheres. The difference in the richness of major microbial groups of archaea between the North and South hemispheres. The significant difference was examined using Wilcoxon rank sum tests. The data originated from global metagenomic data. **Figure S29:** Mean values for soil biodiversity in two hemispheres. The difference in the richness of major microbial groups of bacteria between the North and South hemispheres. The significant difference was examined using Wilcoxon rank sum tests. The data originated from global metagenomic data. **Figure S30:** Latitudinal patterns in soil biodiversity across different continental regions. The latitudinal patterns in the richness of the major bacterial groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data originated from global amplicon data. **Figure S31:** Latitudinal patterns in soil biodiversity across different continental regions. The latitudinal patterns in the richness of the major fungal groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data originated from global amplicon data. **Figure S32:** Latitudinal patterns in soil biodiversity across different continental regions. The latitudinal patterns in the richness of the major protist groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data originated from global amplicon data. **Figure S33:** Latitudinal patterns in soil biodiversity across different continental regions. The latitudinal patterns in the richness of the major invertebrate groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data originated from global amplicon data. **Figure S34:** Latitudinal patterns in soil biodiversity across different continental regions. The latitudinal patterns in the richness of the major bacterial groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data originated from global metagenomic data. **Figure**

S35: Latitudinal patterns in soil biodiversity across different continental regions. The latitudinal patterns in the richness of the major archaeal groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data originated from global metagenomic data. **Figure S36:** Relationship between beta diversity and absolute latitude. Scatter points represent beta diversity values for different taxa. Curves show fitted relationships from ordinary least squares regression including a quadratic term. The data originated from global amplicon data. **Figure S37:** Relationship between beta diversity and absolute latitude. Scatter points represent beta diversity values for different taxa. Curves show fitted relationships from ordinary least squares regression including a quadratic term. The data originated from global metagenomic data. **Figure S38:** Relationship between beta diversity and absolute latitude across continental regions. Curves show fitted relationships from ordinary least squares regression including a quadratic term. The data originated from global amplicon data. **Figure S39:** Estimated coefficients of the latitude–beta diversity relationships were derived from ordinary least squares regression including a quadratic term. Points represent the effect size (estimates) for the quadratic term of latitude, with horizontal bars indicating 95% confidence intervals. The red dashed line indicates zero effect. Asterisks denote significance levels (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). The data originated from global amplicon data. **Figure S40:** Relationship between beta diversity and absolute latitude across continental regions. Curves show fitted relationships from ordinary least squares regression including a quadratic term. The data originated from global metagenomic data. **Figure S41:** Estimated coefficients of the latitude–beta diversity relationships were derived from ordinary least squares regression including a quadratic term. Points represent the effect size (estimates) for the quadratic term of latitude, with horizontal bars indicating 95% confidence intervals. The red dashed line indicates zero effect. Asterisks denote significance levels (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). The data originated from global metagenomic data. **Figure S42:** Changes in soil microbial beta diversity with latitude in the two hemispheres. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. The data originated from global amplicon data. **Figure S43:** Changes in soil microbial beta diversity with latitude in the two hemispheres. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. Significance levels are denoted as follows: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. The data originated from global metagenomic data. **Figure S44:** Latitudinal patterns in soil microbial beta diversity across different continental regions. The latitudinal patterns of microbial beta diversity of the major bacterial groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data originated from global amplicon data. **Figure S45:** Latitudinal patterns in soil microbial beta diversity across different continental regions. The latitudinal patterns of microbial beta diversity of the major fungal groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data

originated from global amplicon data. **Figure S46:** Latitudinal patterns in soil microbial beta diversity across different continental regions. The latitudinal patterns of microbial beta diversity of the major protist groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data originated from global amplicon data. **Figure S47:** Latitudinal patterns in soil microbial beta diversity across different continental regions. The latitudinal patterns of microbial beta diversity of the major invertebrate groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data originated from global amplicon data. **Figure S48:** Latitudinal patterns in soil microbial beta diversity across different continental regions. The latitudinal patterns of microbial beta diversity of the major bacterial groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data originated from global metagenomic data. **Figure S49:** Latitudinal patterns in soil microbial beta diversity across different continental regions. The latitudinal patterns of microbial beta diversity of the major archaeal groups across different latitudinal regions. To ascertain whether latitudinal patterns in soil microbial diversity held consistent across different latitudinal regions, we divided the data into three categories: data from America, data from Africa and Europe, and data from Asia and Australia. Data were fitted using both linear and quadratic models. Linear regressions were shown with dashed lines, and quadratic regressions were shown with solid lines. Grey bands indicate 95% confidence intervals. The data originated from global metagenomic data. **Figure S50:** Microbial beta diversity in the North and South hemispheres. The significant difference was examined using Wilcoxon rank sum tests. The data originated from global amplicon data. **Figure S51:** Microbial beta diversity in the North and South hemispheres. The significant difference was examined using Wilcoxon rank sum tests. The data originated from global metagenomic data. **Figure S52:** Difference of beta diversity components for taxa across multiple trophic levels. The data originated from global amplicon data. **Figure S53:** Difference of beta diversity components for taxa across multiple trophic levels. The data originated from global metagenomic data. **Figure S54:** Relationship between beta diversity components and absolute latitude. Curves show fitted relationships from ordinary least squares regression including a quadratic term. The data originated from global amplicon data. **Figure S55:** Relationship between beta diversity components and absolute latitude. Curves show fitted relationships from ordinary least squares regression including a quadratic term. The data originated from global metagenomic data. **Figure S56:** The independent explanatory power of latitude in driving species richness variations. Both variation partitioning analysis (VPA) and hierarchical partitioning were applied. **Figure S57:** The independent explanatory power of latitude in driving community turnover. The specific contributions of latitude to beta diversity were assessed via Partial Mantel tests, while controlling for covariation with other environmental factors. **Figure S58:** Variance partitioning analysis (VPA) reveals the contributions of key environmental categories (climatic conditions, vegetation, spatial, and soil properties) influencing species richness. The data originated from global amplicon (a) and metagenomic (b) data. **Figure S59:** Spearman correlations between environmental predictors and multitrophic biodiversity, with

insignificant correlations left blank. D_{equator}, distance from the equator; MAT, mean annual temperature; TSEA, temperature seasonality; PSEA, precipitation seasonality; MAP, mean annual precipitation; Soil C, soil carbon; Soil N, soil nitrogen; Soil C/N, soil carbon: nitrogen ratio; Texture, clay + silt %. The data originated from global amplicon (a) and metagenomic (b) data. **Figure S60:** Comparison of standardized path coefficients among taxonomic groups, evaluated by multi-group structural equation models. Each segment represents the strength and direction of a specific path within a group. Solid and dashed lines indicate significant and non-significant paths, respectively ($*p < 0.05$; $**p < 0.01$; $***p < 0.001$). **Figure S61:** The predicted distribution of microbial richness across the globe (pixel size of 0.10×0.10 degrees; 10 km resolution), based on spatially explicit random forest models. Areas in grey are those regions with excluded land cover types (e.g., glaciers, desert). Data were standardized to a value between 0 and 1. The data originated from global amplicon data. **Figure S62:** The maps of uncertainty (5% confidence interval) of multitrophic soil biodiversity across the globe. The uncertainty of the predictions was assessed using the Quantile Regression Forests methods in quantregForest R package. The data originated from global amplicon data. **Figure S63:** The maps of uncertainty (95% confidence interval) of multitrophic soil biodiversity across the globe. The uncertainty of the predictions was assessed using the Quantile Regression Forests methods in quantregForest R package. The data originated from global amplicon data. **Figure S64:** The predicted distribution of microbial richness across the globe (pixel size of 0.10×0.10 degrees; 10 km resolution), based on spatially explicit random forest models. Areas in grey are those regions with excluded land cover types (e.g., glaciers, desert). Data were standardized to a value between 0 and 1. The data originated from global metagenomic data. **Figure S65:** The maps of uncertainty (5% confidence interval) of multitrophic soil biodiversity across the globe. The uncertainty of the predictions was assessed using the Quantile Regression Forests methods in quantregForest R package. The data originated from global metagenomic data. **Figure S66:** The maps of uncertainty (95% confidence interval) of multitrophic soil biodiversity across the globe. The uncertainty of the predictions was assessed using the Quantile Regression Forests methods in quantregForest R package. The data originated from global metagenomic data. **Table S1:** The initial model of the multi-group structural equation model. Soil C, soil carbon; Soil N, soil nitrogen; NPP, net primary productivity. **Table S2:** Model comparison results of alpha diversity. For each microbial group, the table presents: coefficient estimate for the linear latitude term, lower and upper bounds of the 95% confidence interval, p value, significance indication ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$; ns = not significant), AIC value, and theta parameter (dispersion parameter of the negative binomial distribution). The data originated from global amplicon data. **Table S3:** Model comparison results of alpha diversity. For each microbial group, the table presents: coefficient estimate for the linear latitude term, lower and upper bounds of the 95% confidence interval, p value, significance indication ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$; ns = not significant), AIC value, and theta parameter (dispersion parameter of the negative binomial distribution). The data originated from global metagenomic data. **Table S4:** Representation of major soil taxa in both datasets. This table shows the mean relative abundance and detection frequency of the 51 analysed taxa. **Table S5:** Model comparison results of gamma diversity. For each microbial group, the table presents: coefficient estimate for the linear latitude term, lower and upper bounds of the 95% confidence interval, p value, significance indication ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$; ns = not significant), AIC value, and theta parameter (dispersion parameter of the negative binomial distribution). The data originated from global amplicon data. **Table S6:** Model comparison results of gamma diversity. For each microbial group, the table presents: coefficient estimate for the linear latitude term, lower and upper bounds of the 95% confidence interval, p value, significance indication ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$; ns = not significant), AIC value, and theta parameter (dispersion parameter of the negative binomial distribution). The data originated from global metagenomic data. **Table S7:** Model comparison results of alpha diversity across latitudinal regions. For each microbial group, the table presents: coefficient estimate for the linear latitude term, lower and upper bounds of the 95% confidence interval, p value, significance

indication ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$; ns = not significant), AIC value, and theta parameter (dispersion parameter of the negative binomial distribution). The data originated from global amplicon data. **Table S8:** Model comparison results of alpha diversity across latitudinal regions. For each microbial group, the table presents: coefficient estimate for the linear latitude term, lower and upper bounds of the 95% confidence interval, p value, significance indication ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$; ns = not significant), AIC value, and theta parameter (dispersion parameter of the negative binomial distribution). The data originated from global metagenomic data. **Table S9:** Model comparison results of gamma diversity across latitudinal regions. For each microbial group, the table presents: coefficient estimate for the linear latitude term, lower and upper bounds of the 95% confidence interval, p value, significance indication ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$; ns = not significant), AIC value, and theta parameter (dispersion parameter of the negative binomial distribution). The data originated from global amplicon data. **Table S10:** Model comparison results of gamma diversity across latitudinal regions. For each microbial group, the table presents: coefficient estimate for the linear latitude term, lower and upper bounds of the 95% confidence interval, p value, significance indication ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$; ns = not significant), AIC value, and theta parameter (dispersion parameter of the negative binomial distribution). The data originated from global metagenomic data. **Table S11:** Model comparison results of beta diversity. For each microbial group, the table presents: coefficient estimate for the linear latitude term, lower and upper bounds of the 95% confidence interval, p value, significance indication ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$; ns = not significant), and AIC value. The data originated from global amplicon data. **Table S12:** Model comparison results of beta diversity. For each microbial group, the table presents: coefficient estimate for the linear latitude term, lower and upper bounds of the 95% confidence interval, p value, significance indication ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$; ns = not significant), and AIC value. The data originated from global metagenomic data. **Table S13:** Spatial autocorrelation analyses using Moran's I with Monte Carlo permutation tests (999 permutations) on the observed species richness. $**p < 0.01$; $*p < 0.05$; ns, no significant. **Table S14:** Results of the model selection procedure for influencing microbial richness. This analysis was conducted using the site as a random effect (random intercept, 1|site). We indicate marginal and conditional R^2 , the number of observations (N), predictor estimates, standard errors, p values, the number of times each predictor was selected in the set of best models (n), predictor importance based on the sum of weights (sum of weights), variance inflation factors (VIF), and the results of Moran tests for spatial autocorrelation. These tests were performed with the residuals of the models at different spatial scales (using the 10, 20, and 50 closest plots); their results show no evidence of spatial autocorrelation. The VIF values obtained were below 10 in all cases; hence, multicollinearity was not problematic. MAT, mean annual temperature; TSEA, temperature seasonality; PSEA, precipitation seasonality; MAP, mean annual precipitation; Soil C, soil carbon; Soil N, soil nitrogen; C/N, carbon: nitrogen ratio; Texture, clay + silt %; NPP, net primary productivity. The data originated from global amplicon data. **Table S15:** Results of the model selection procedure for influencing microbial richness. This analysis was conducted using the site as a random effect (random intercept, 1|site). We indicate marginal and conditional R^2 , the number of observations (N), predictor estimates, standard errors, p values, the number of times each predictor was selected in the set of best models (n), predictor importance based on the sum of weights (sum of weights), variance inflation factors (VIF), and the results of Moran tests for spatial autocorrelation. These tests were performed with the residuals of the models at different spatial scales (using the 10, 20, and 50 closest plots); their results show no evidence of spatial autocorrelation. The VIF values obtained were below 10 in all cases; hence, multicollinearity was not problematic. MAT, mean annual temperature; TSEA, temperature seasonality; PSEA, precipitation seasonality; MAP, mean annual precipitation; Soil C, soil carbon; Soil N, soil nitrogen; C/N, carbon: nitrogen ratio; Texture, clay + silt %; NPP, net primary productivity. The data originated from global metagenomic data. **Table S16:** The variance inflation factor (VIF) values for each variable. Soil C, soil carbon; Soil N, soil nitrogen; NPP, net primary productivity. **Table S17:** The

measurement invariance across groups at four hierarchical levels: configurational, metric, structural, and strict invariance. **Table S18:** The average standardized path coefficients for all taxonomic groups. Soil C, soil carbon; Soil N, soil nitrogen; NPP, net primary productivity. $***p < 0.001$; $**p < 0.01$; $*p < 0.05$. **Table S19:** Parameters of random forest model for each main group. mtry, number of variables randomly sampled in constructing decision tree; RMSE, root mean square error; MAE, mean absolute error. The data originated from global amplicon data. **Table S20:** Parameters of random forest model for each main group. mtry, number of variables randomly sampled in constructing decision tree; RMSE, root mean square error; MAE, mean absolute error. The data originated from global metagenomic data. **Table S21:** The importance of each environmental variable in predicting microbial richness. This was analysed by random forest models. MAT, mean annual temperature; TSEA, temperature seasonality; PSEA, precipitation seasonality; MAP, mean annual precipitation; Soil C, soil carbon; Soil N, soil nitrogen; C/N, Soil carbon: nitrogen ratio; Texture, clay + silt %. The data originated from global amplicon data. **Table S22:** The importance of each environmental variable in predicting microbial richness. This was analysed by random forest models. MAT, mean annual temperature; TSEA, temperature seasonality; PSEA, precipitation seasonality; MAP, mean annual precipitation; Soil C, soil carbon; Soil N, soil nitrogen; C/N, Soil carbon: nitrogen ratio; Texture, sand %. The data originated from global metagenomic data.